Energy Reviews 5 (2026) 100188

Contents lists available at ScienceDirect

Energy Reviews

journal homepage: www.journals.elsevier.com/energy-reviews

A review on in-situ methanation based on CO_2/H_2O Co-electrolysis in solid oxide electrolytic cells addressing the thermo-kinetic mismatch

Feiyang Liao [a,b], Junbo Xu [a], Chao Yang [a,c], Wei Zhou [b,d,*]

[a] State Key Laboratory of Biochemical Engineering, Institute of Process Engineering, Chinese Academy of Sciences, Beijing, 100190, China
[b] Suzhou Laboratory, Suzhou, 215000, China
[c] School of Chemical Engineering, University of Chinese Academy of Sciences, Beijing, 100049, China
[d] State Key Laboratory of Materials-Oriented Chemical Engineering, College of Chemical Engineering, Nanjing Tech University, Nanjing, 211816, China

ARTICLE INFO

Keywords:
Renewable fuel
Co-electrolysis
SOEC
Power-to-methane
Thermo-kinetic mismatch
Thermo-electric coupling

ABSTRACT

Solid oxide electrolysis cells (SOECs) offer a compelling platform for converting renewable electricity and captured CO_2 into methane by integrating CO_2/H_2O co-electrolysis with in-situ methanation. This single-reactor concept promises process intensification, superior heat integration, and compatibility with existing natural gas infrastructure. However, its development is fundamentally constrained by the thermo-kinetic mismatch: efficient co-electrolysis typically requires temperatures above 600 °C to minimize ohmic and polarization losses, whereas methanation is thermodynamically favored below 400 °C. This review examines recent progress in direct methane synthesis within SOECs from the perspective of mitigating this mismatch systematically. We first clarify the reaction pathways, which including CO_2/H_2O electroreduction, reverse water-gas shift, methanation, carbon formation, and electrochemical promotion, and discuss the reaction mechanism of the direct methane synthesis within SOECs. Subsequently, we analyze material design strategies with emphasis on multifunctional cathodes, integration of methanation-active catalysts, and the enabling roles of oxygen electrodes and electrolytes in achieving medium-temperature operation. Reactor engineering approaches, ranging from single-temperature-zone to two-temperature-zone and thermally intensified configurations, are then evaluated to illustrate how spatial separation and flow-field regulation can enhance methane selectivity and overall efficiency. Finally, we identify key challenges and future directions in mechanistic understanding, integrated materials design, reactor optimization, and long-term durability. This review highlights that coordinated advances across reaction mechanism, thermo-electrocatalytic materials, and reactor engineering are essential to translate SOEC in-situ methanation from laboratory demonstration into practical power-to-methane applications.

1. Introduction

Fossil resources are being depleted at an accelerating rate, and carbon-based energy systems are imposing a heavier environmental burden; together, these pressures have pushed renewable energy technologies to the center of attention. Renewable electricity is widely seen as cleaner and more sustainable, yet large-scale deployment still runs into familiar obstacles: Its regionality, intermittency, and dynamism. For that reason, converting renewable electricity into chemical fuels has become a key route for both energy storage and carbon utilization.

Among the available conversion technologies, solid oxide electrolysis cells (SOECs) have drawn particular interest [1–6]. The reason is straightforward: they offer high energy-conversion efficiency, strong fuel flexibility, and the ability to turn both electrical and thermal energy into chemical energy in the form of H_2, CO, syngas, and even hydrocarbons. Viewed through the Power-to-X framework, CO_2/H_2O co-electrolysis in SOECs is especially attractive because it provides a direct pathway for converting captured CO_2 and renewable electricity into carbon-containing fuels. In oxygen-ion-conducting SOECs (Fig. 1), CO_2 and H_2O enter the fuel electrode, where they are electrochemically reduced to CO and H_2; at the same time, O^{2-} ions migrate through the dense electrolyte and release oxygen at the oxygen electrode. Under these high-temperature conditions, electrochemical reduction is accompanied by rapid heterogeneous reactions, resulting in a strongly coupled reaction environment involving charge transfer, surface catalysis, and mass transport.

* Corresponding author. Suzhou Laboratory, Suzhou, 215000, China.
E-mail address: zhouwei1982@njtech.edu.cn (W. Zhou).

https://doi.org/10.1016/j.enrev.2026.100188
Received 6 April 2026; Received in revised form 2 May 2026; Accepted 14 May 2026
2772-9702/© 2026 The Authors. Published by Elsevier Ltd on behalf of State Key Laboratory of Intelligent Construction and Healthy Operation and Maintenance of Deep Underground Engineering, Shenzhen University. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

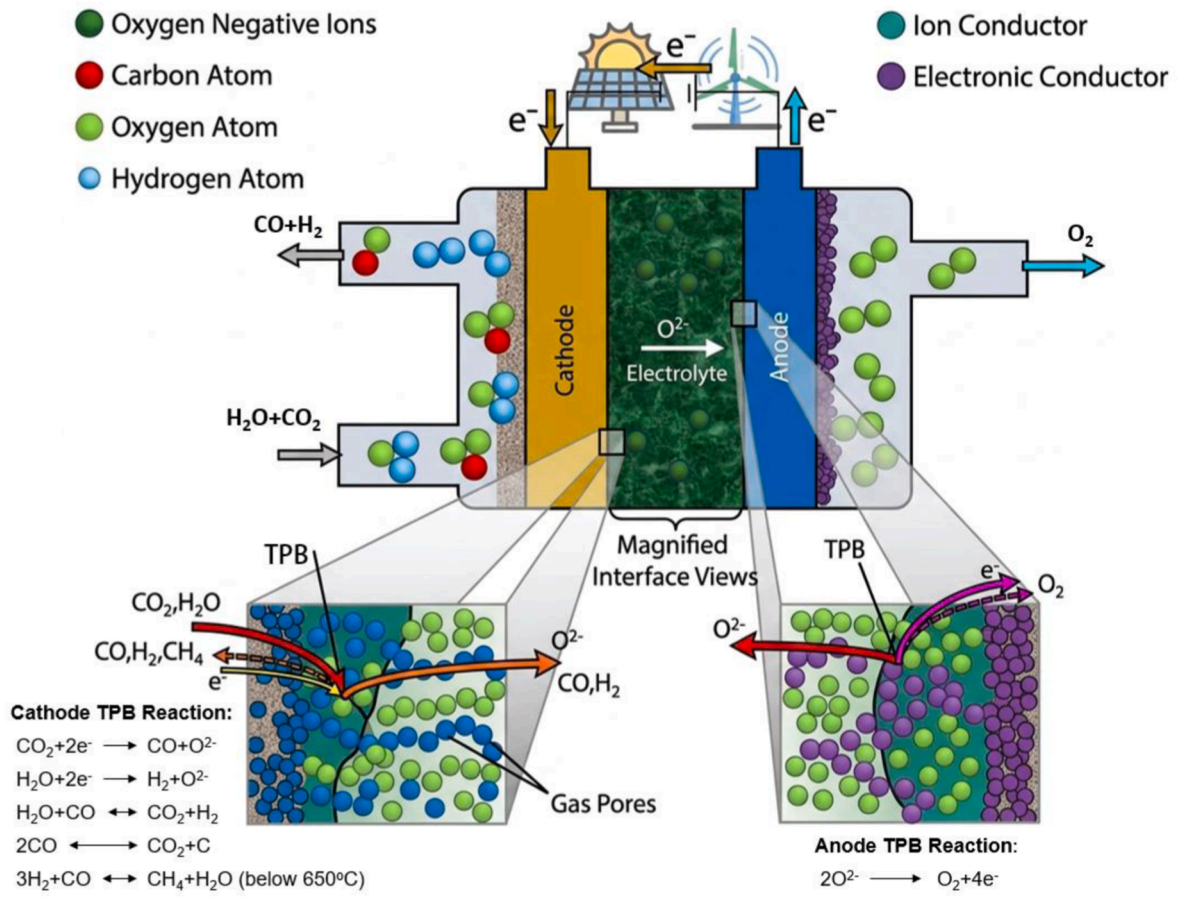

Fig. 1. The schematic diagram of the working principle of O-SOEC co-electrolysis CO_2/H_2O.

Among the possible products, methane is of particular interest due to its high volumetric energy density, compatibility with existing natural gas infrastructure, and strategic value for long-duration and seasonal energy storage. Methane synthesis based on SOEC co-electrolysis can proceed through two distinct routes (Fig. 2). For two-step thermally decoupled route, the SOEC mainly serves as a source of H_2 or syngas, which is subsequently converted to methane in a separate methanation reactor. This route has the potential to reduce the high electricity demand of electrolysis by recovering waste heat from the exothermic methanation reaction. In contrast, the one-step thermally coupled route

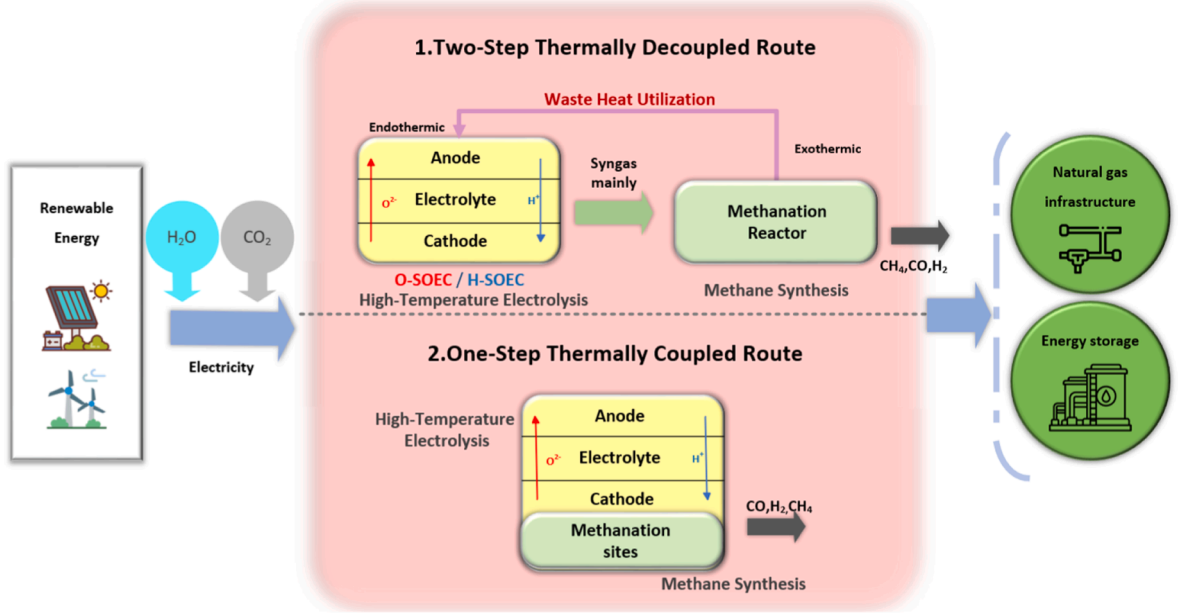

Fig. 2. Schematic illustration of two-step thermally decoupled and one-step thermally coupled methane synthesis routes based on SOECs.

produces methane in situ within the SOEC reactor, thereby eliminating the need for a downstream methanation unit and creating the possibility of tighter thermal integration, higher overall efficiency, and more compact system design [7–9]. This review focuses on the latter one, namely SOEC in-situ methanation.

Depending on the electrolyte type, SOECs can be classified as proton-conducting SOECs (H-SOECs) and oxygen-ion-conducting SOECs (O-SOECs). H-SOECs offer advantages at relatively lower temperatures because of the lower activation energy and higher conductivity of proton transport, which is potentially favorable for methane synthesis [10,11]. However, uncertainties in CO_2 conversion pathways, limited Faradaic efficiency and anode stability during co-electrolysis, and the insufficient chemical stability of proton-conducting electrolytes under CO_2-containing conditions have so far restricted their development [12–14]. Consequently, most current studies on direct methane synthesis are still centered on O-SOEC systems.

The development of SOEC in-situ methanation has progressed steadily from early proof-of-concept studies to the substantial methane yields achieved today [15–19]. And the SOEC in-situ methanation is not simply the sequential combination of electrolysis and methanation, but the outcome of a strongly coupled network involving H_2O electrolysis, possible direct CO_2 electroreduction, reverse water-gas shift (RWGS), thermal catalytic hydrogenation of CO and CO_2, and side reactions such as carbon deposition [20–25] (Fig. 3).

A central challenge arising from in-situ methanation based on CO_2/H_2O co-electrolysis in SOEC is the thermo-kinetic mismatch between the two main functionalities of the process. In SOECs, the total energy input required for electrolysis (ΔH) remains nearly unchanged with increasing temperature, while the electrical energy demand (ΔG) decreases and the thermal energy requirement increases accordingly. This behavior is observed for both H_2O and CO_2 electrolysis (Fig. 4a), and is more pronounced in CO_2/H_2O co-electrolysis (Fig. 4b), where the reduction in electrical energy is approximately compensated by the increase in thermal energy. Given that thermal energy is generally less costly than electrical energy, high-temperature operation is thermodynamically favorable for CO_2/H_2O co-electrolysis. Moreover, elevated temperature lowers ohmic resistance and promotes electrode reaction kinetics. Therefore, high-temperature SOEC operation is generally considered advantageous for CO_2/H_2O co-electrolysis in terms of both energy efficiency and operating cost. The curves shown in Fig. 4c illustrate the relationship between ΔG and ΔH for the water gas shift reaction (WGS). Notably, at 816 °C, the ΔG value for the WGS reaction reaches zero, indicating the WGS reaction's preference at lower temperatures (ΔG<0), while the opposite holds true for the reverse process (RWGS). In contrast, according to the thermodynamic diagram of CO methanation (Fig. 5), the CH_4 yield reaches nearly 100% at pressures of 1 atm or higher and temperatures below 250 °C, and the optimal temperature range for the reaction, considering both thermodynamic and kinetic aspects, falls between 250 °C and 350 °C. CO_2 methanation can achieve nearly 100% CH_4 selectivity when the pressure is maintained at 1 atm or above and the temperature is below 400 °C. Compared with CO methanation under similar temperature and pressure conditions, however, CO_2 methanation is generally more difficult, mainly because CO_2 is a highly stable molecule and its activation requires more demanding catalytic conditions. As shown in Fig. 6, the favorable temperature window for CO_2 methanation is typically located between 250 °C and 400 °C. This temperature preference raises an important question for SOEC-based in-situ methanation: how can a process that benefits from high-temperature electrolysis also maintain conditions suitable for methane formation? At temperatures above 700 °C, the enhanced electrolysis kinetics and intensified RWGS activity tend to drive the system toward syngas production rather than CH_4 formation [22,23]. When the temperature decreases below 400 °C, the equilibrium shifts toward methane [25], but the SOEC may suffer from increased ohmic resistance and polarization losses. A more realistic strategy is therefore to operate, whenever possible, in an intermediate temperature range of approximately 500–650 °C [26–28]. When medium-temperature operation is not feasible, spatial separation of electrolysis and methanation zones becomes essential.

To address this limitation, two main reactor-operation modes have been explored [8] (Fig. 7). In the single-temperature-zone mode, the cathode itself acts as the methanation-active interface or is directly coupled with a catalyst layer [26]. In the two-temperature-zone mode, electrolysis and methanation are spatially separated, and methane is formed in a cooler catalyst bed located near or downstream of the electrode [29,30]. In addition to temperature, methane yield is strongly influenced by pressure, applied potential, feed composition, and flow configuration, indicating that SOEC in-situ methanation is governed by the coupled effects of materials, operating conditions, and reactor architecture [17,19,31–34].

Long-term durability, however, remains one of the main barriers that must be overcome before SOEC in-situ methanation can move toward practical deployment. In recent years, considerable progress has been made in improving electrode activity, electrolyte conductivity, and interfacial stability, yet the degradation rates reported for even relatively stable SOEC stacks are still higher than those required for commercial operation [35]. Despite these unresolved durability issues, industrial and engineering interest in SOEC-based methane synthesis is

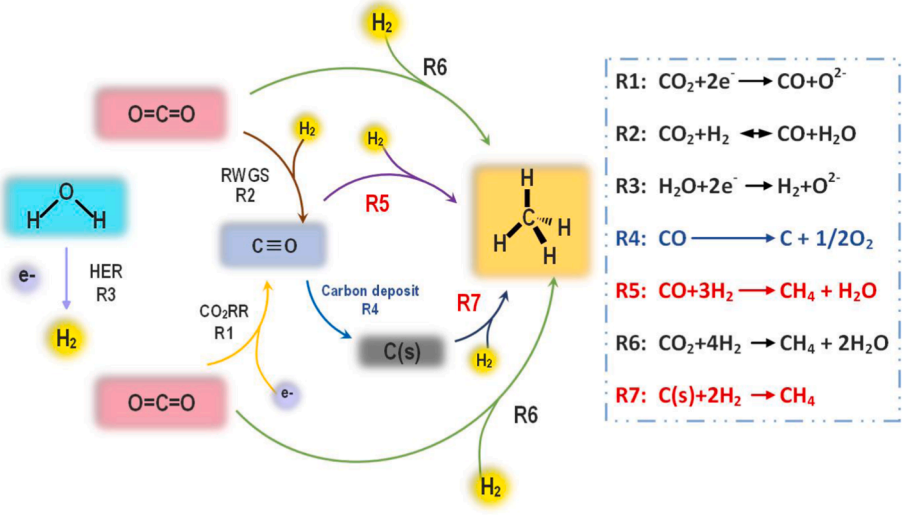

Fig. 3. The reaction path of CO_2/H_2O co-electrolysis to produce methane by O-SOEC.

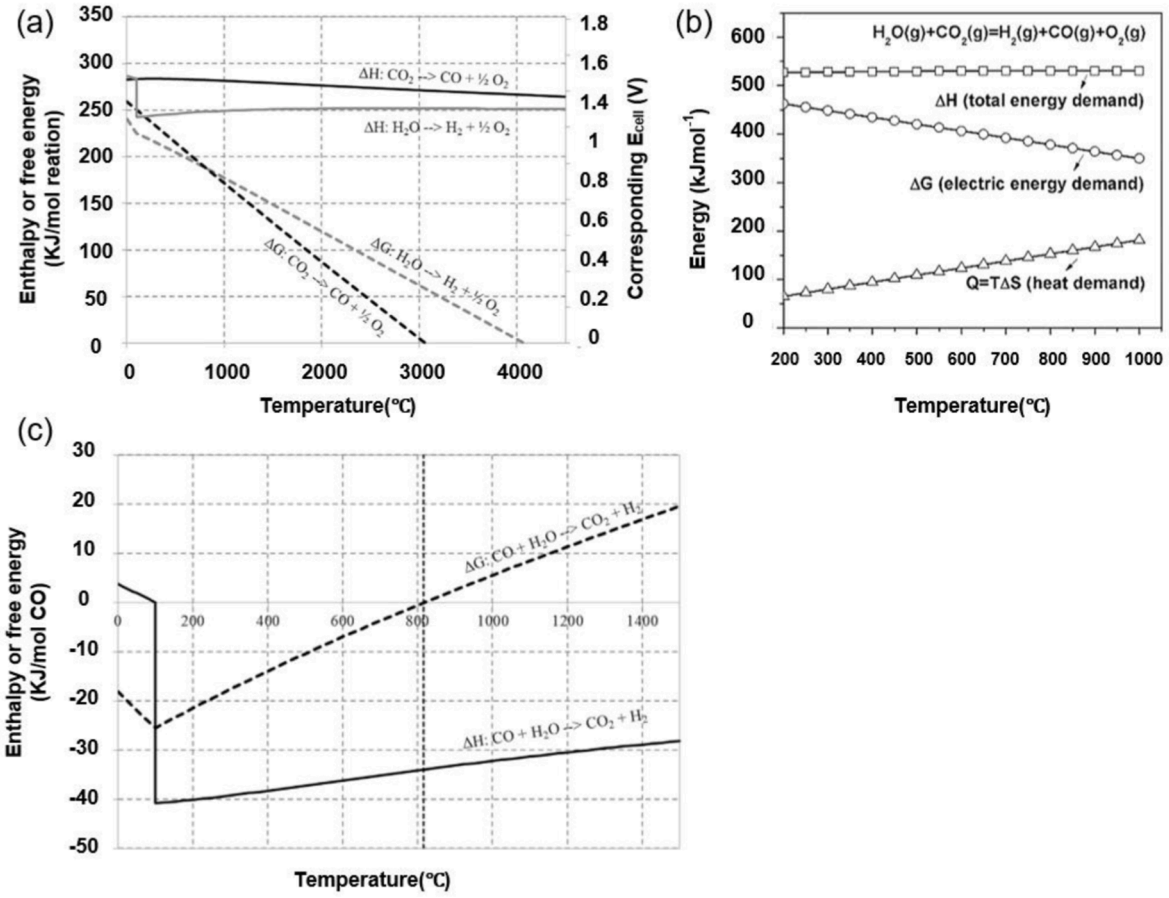

Fig. 4. (a) Enthalpy and free energy of reaction for the reduction of both H_2O and CO_2 [23]; (b) the temperature dependency of the co-electrolysis thermodynamics [22]; (c) enthalpy and free energy of reaction for the water gas shift reaction [23].

increasing. Recent demonstrations of high-efficiency methane production under hydrogen-rich conditions, modular reactor concepts with scalable output, and dedicated commercialization-oriented development programs all suggest that the technology is no longer limited to proof-of-concept studies [36,37].

Against this background, a systematic review of SOEC in-situ methanation is both necessary and timely. Current research is progressing along several closely connected directions, including the clarification of reaction pathways, the design of multifunctional materials that can support both CO_2/H_2O co-electrolysis and methane formation, and reactor engineering strategies aimed at alleviating the thermo-kinetic mismatch through local control of temperature, flow field, and reactive atmosphere [38]. Following this logic, this review first discusses the reaction mechanisms involved in CO_2/H_2O co-electrolysis and methane formation in SOECs. It then examines material design strategies for direct methane synthesis, with particular attention to cathode functionality and electrolyte/electrode compatibility. Reactor design approaches for improving thermo-electrical coupling are subsequently analyzed, before the review concludes with remaining challenges and future perspectives.

2. Reaction mechanisms of direct methane synthesis via CO_2/H_2O co-electrolysis in SOEC

The working principle outlined above reveals the intricate reaction and transfer processes involved in the SOECs during the CO_2/H_2O co-electrolysis reaction. At the fuel electrode's triple-phase boundary (TPB), a charge transfer reaction occurs, releasing ions that are subsequently transported via the ion conductor to the oxygen electrode. Concurrently, heterogeneous catalytic reactions, including the RWGS, methanation, and carbon deposition, take place on the surface of the nickel metal conductor. These reactions, along with the bulk diffusion processes within the porous electrode, collectively influence the electrochemical performance of the SOEC's porous fuel electrode.

The lack of adequate in situ techniques at elevated operating temperatures (500–1000 °C) hinders direct observation of surface reactions [39,40]. Nevertheless, a combination of experimental and computational studies has enabled the proposal of several mechanistic pathways. This chapter first describes the individual reduction mechanisms of CO_2 and H_2O, methane formation, then discusses the coupling reactions, and finally integrates these into a comprehensive reaction network for in-situ methane formation.

- CO_2 electrochemical reduction

Limited insights exist regarding the electrolysis mechanism of CO_2 on the fuel side. Although the electrochemical reduction of CO_2 in SOECs occurs on different cathode materials, several common mechanistic features can be identified (Table 1).

In conventional Ni-YSZ electrodes, the adsorbed CO_2 species are activated mainly on metallic Ni sites, where direct electron transfer facilitates C=O bond cleavage and ultimately leads to CO formation (Fig. 8a), and this pathway is relatively straightforward. The situation becomes more complex when oxide cathodes are involved. For perovskite-type materials such as LSF, spectroscopic and computational studies suggest that CO_2 activation often proceeds through carbonate-like CO_3^{2-} intermediates (Fig. 8b). Such intermediates indicate a strong interaction between CO_2 molecules and surface lattice oxygen,

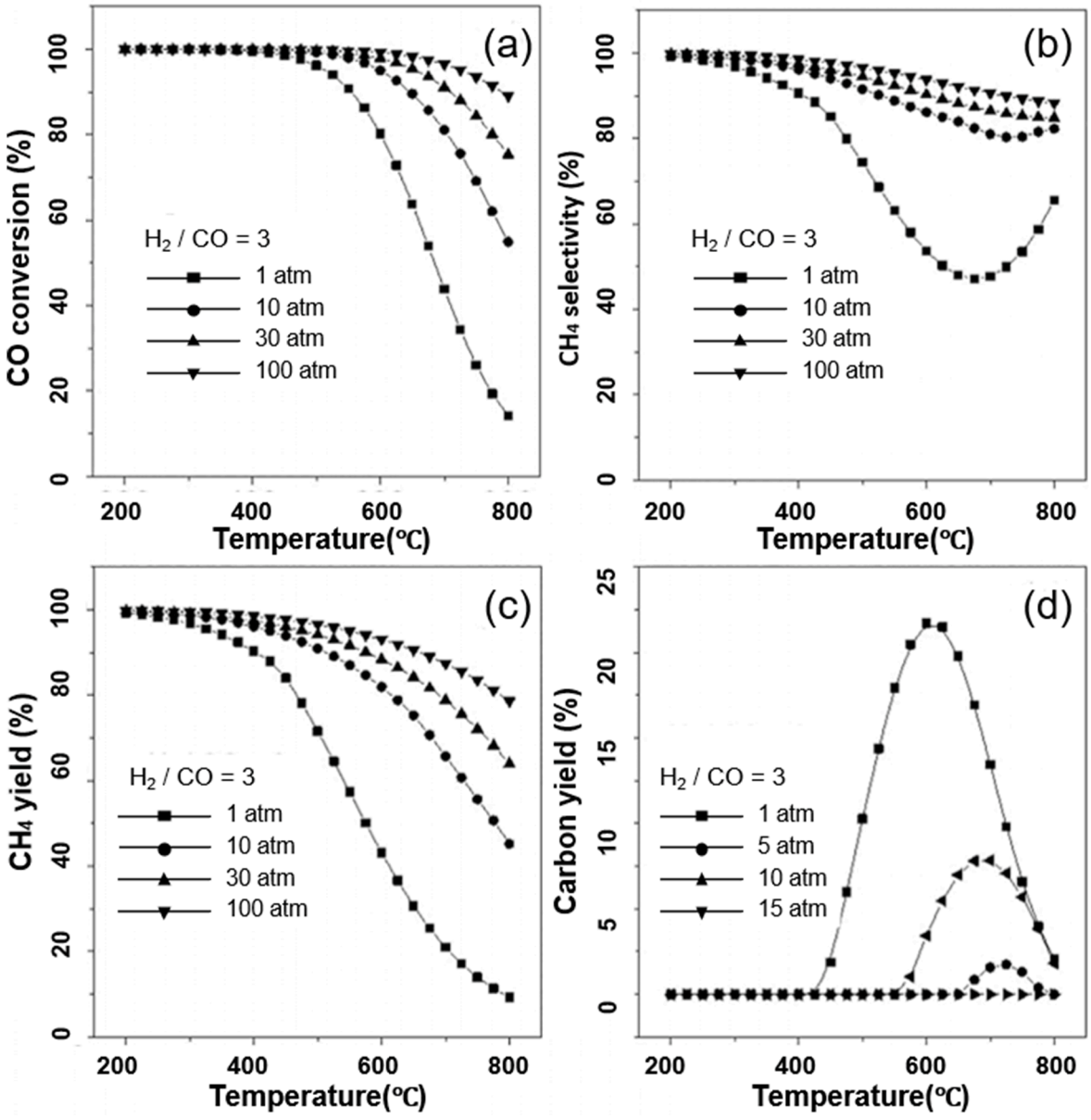

Fig. 5. Results of CO methanation reactions under different temperature and pressure conditions; (a) CO conversion; (b) CH_4 selectivity; (c) CH_4 yield; (d) Carbon yield [25].

suggesting that surface oxygen species participate directly in bond activation rather than simply providing a passive structural environment. Ceria-based electrodes, represented by SDC, introduce another layer of complexity. In these systems, the Ce^{4+}/Ce^{3+} redox couple and the abundance of oxygen vacancies strongly affect CO_2 adsorption, activation, and subsequent conversion. CO_2 reduction is therefore not controlled by electron transfer alone; it also depends on vacancy formation and migration, as well as the stability of carbonate-type surface species (Fig. 8c). From a practical viewpoint, this coupling between redox chemistry and defect chemistry is particularly relevant because it influences both catalytic activity and resistance to carbon-related degradation. Skafte et al. [41] further identified an inhibition mechanism on ceria, providing useful evidence that the surface properties of ceria and other vacancy-rich oxides can be tuned to improve carbon tolerance. As a result, doped ceria has become a model system for probing the mechanistic complexity of practical electrochemical reactions.

- H_2O electrolysis mechanism

Compared with CO_2 electrolysis, whose mechanistic interpretation is still not fully consistent across different studies, the reaction mechanism of H_2O electrolysis in SOECs has been investigated more extensively and is now relatively better established. H_2O electrolysis mechanisms are largely influenced by the electrode microstructure and the catalytic properties of the electrolyte interface. However, these factors do not mainly fall within the scope of our research focus on addressing the thermo-kinetic mismatch, and the discussion of the H_2O electrolysis mechanism specifically focuses on a single representative system, namely Ni-YSZ-based cermet electrodes, which remain the most widely studied and practically applied SOEC cathodes, offering a well-characterized benchmark for understanding H_2O electrolysis mechanisms. The hydrogen overflow mechanism and the oxygen overflow mechanism are among the most extensively researched and utilized. Below, a simplified description of the hydrogen overflow mechanism is provided.

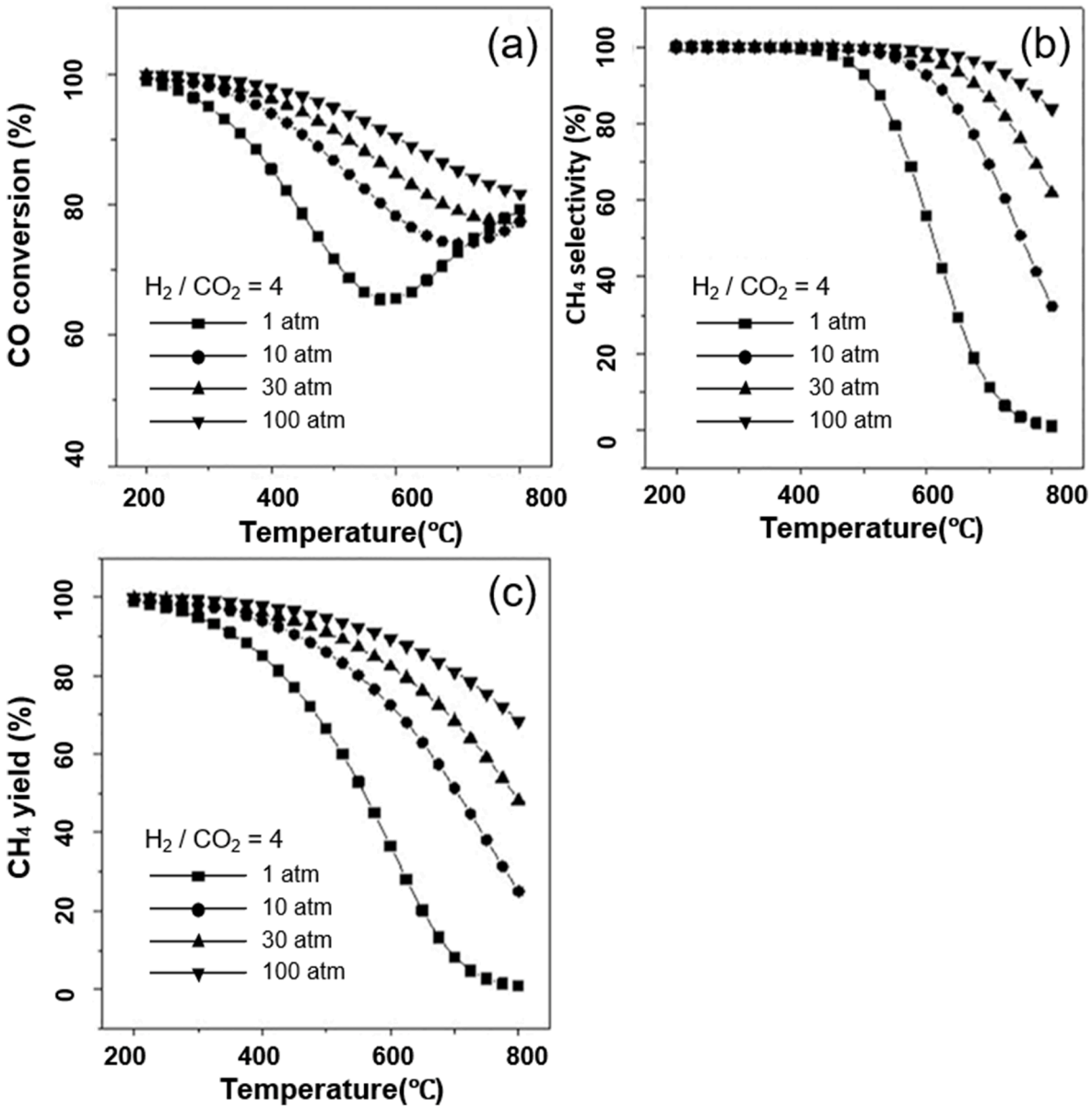

Fig. 6. Results of CO_2 methanation reactions under different temperature and pressure conditions (a) CO_2 conversion; (b) CH_4 selectivity; (c) CH_4 yield [25].

(1) Oxygen transfer on YSZ surface and bulk

$$O^{2-}(YSZ) + V_O^{\cdot\cdot}(YSZ) \Leftrightarrow O_O^X(Ni) + (YSZ)$$

(2) Adsorption/desorption on YSZ surfaces

$$H_2O(g) + (YSZ) \Leftrightarrow H_2O(YSZ)$$

(3) Two-step chain charge transfer reaction at TPB

$$(Ni) + H_2O\,(YSZ) + e^- \Leftrightarrow H\,(Ni) + OH^-(YSZ)$$
$$(Ni) + OH^-(YSZ) + e^- \Leftrightarrow H\,(Ni) + O^{2-}(YSZ)$$

(4) Adsorption/desorption on the Ni surface

$$2H\,(Ni) \Leftrightarrow H_2\,(g) + 2\,(Ni)$$

- Methane formation

During co-electrolysis in SOECs, the generated intermediates—primarily H_2 and CO—serve directly as precursors for subsequent methanation reactions. Hydrogen participates in catalytic hydrogenation of CO and CO_2, while CO can be further converted either through CO methanation or through RWGS to regenerate CO_2 as a reactant. Notebly, In the SOEC co-electrolysis of CO_2/H_2O, the carbon monoxide generated in situ dissociates and accumulates on the cathode surface with increasing applied voltage. Deposited carbon directly participates in the electrochemical reaction of TPB during the direct synthesis of methane via SOEC [44]:

$$CO\,(Ni) + (YSZ) + 2e^- \leftrightarrow C\,(Ni) + O^{2-}\,(YSZ)$$

$$C\,(s) + 2H_2 \rightarrow CH_4$$

Although deposited carbon plays a direct role in the production of methane through SOEC co-electrolysis [27], its role in SOEC systems must be carefully distinguished from other electrocatalytic and thermocatalytic pathways. Carbon deposition, rather than being a constructive mechanism, is primarily regarded as a degradation pathway that adversely affects electrode stability and performance.

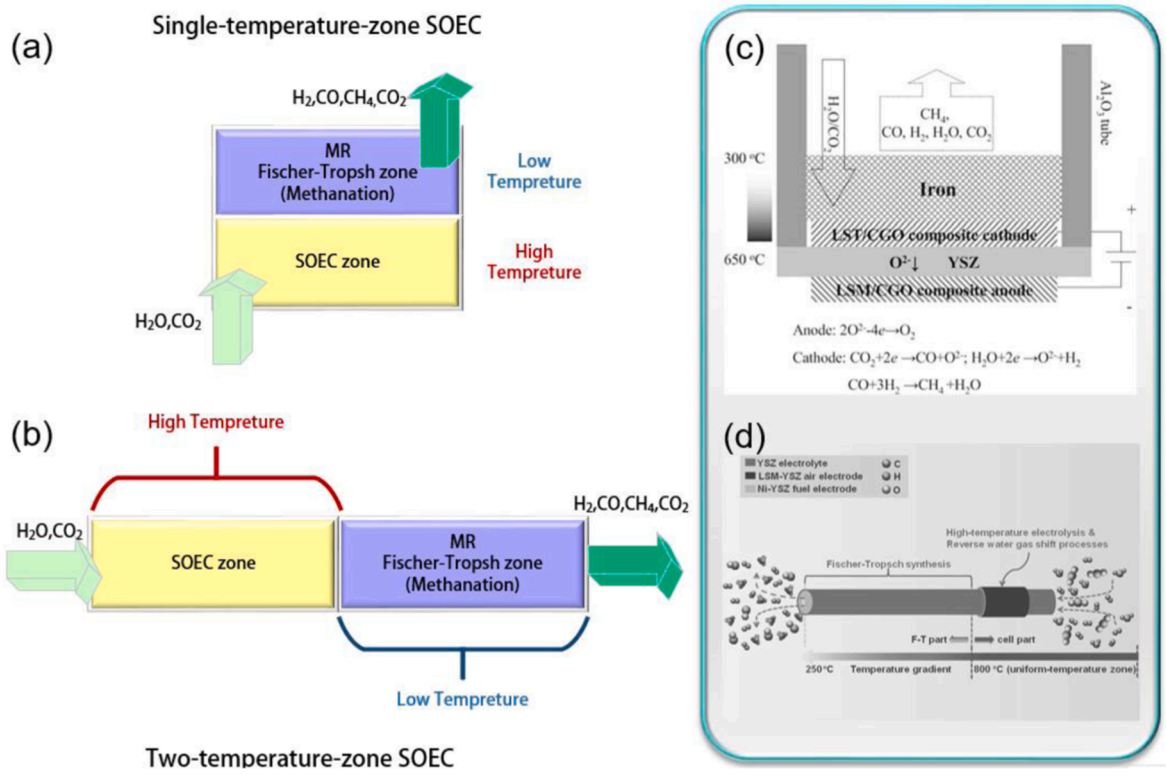

Fig. 7. (a) Single-temperature-zone configuration; (b) two-temperature-zone configuration [8]; (c) typical case of single-temperature-zone [26]; (d) typical case of two-temperature zone [30].

Table 1
Proposed mechanisms of CO_2 electrochemical reduction on typical SOEC cathode materials.

Cathode material	Method	Reaction pathway	Key intermediates/features
Ni-YSZ fuel electrode [41]	Experimental studies and mechanistic inference	(1) $CO_2(g) \rightarrow CO_2^*$ (2) $CO_2^* + 2e^- \rightarrow CO^* + O^{2-}$ (3) $CO^* \rightarrow CO\ (g)$	Adsorbed CO_2 and CO* species
LSF-based cathode [42]	Raman spectroscopy and DFT	(1) $CO_2\ (gas) = CO_2\ (ad)$ (2) $CO_2\ (ad) + O^{2-}\ (lattice) = CO_3^{2-}\ (ad)$ (3) $CO_3^{2-}\ (ad) + e^- = CO_2^{2-}\ (activated\ bent) + O^{2-}\ (latti)$ (4) $CO_2^{2-}\ (activated\ bent) + e^- = CO\ (ad) + O^{2-}\ (ad)$ (5) $CO\ (ad) = CO\ (gas)$	Carbonate (CO_3^{2-}) intermediate, adsorbed CO
SDC [43]	APXPS	(1) $2\left(Ce_{Ce,b}^{\times} + e_M^- \Leftrightarrow Ce_{Ce,b}'\right)$ $V_{O,YSZ}^{\cdot\cdot} \Leftrightarrow V_{O,b}^{\cdot\cdot}$ (2) $2\left(Ce_{Ce,b}' + e_M^- \Leftrightarrow Ce_{Ce,s}'\right)$ $V_{O,b}^{\cdot\cdot} \Leftrightarrow V_{O,s}^{\cdot\cdot}$ (3) $CO_{2(g)} + O_{O,s}^{\times} + Ce_{Ce,s}' \Leftrightarrow (CO_3)_{O,s}' + Ce_{Ce,s}^{\times}$ (4) $(CO_3)_{O,s}' + Ce_{Ce,s}' + V_{O,s}^{\cdot} \Leftrightarrow CO_{(g)} + 2O_{O,s}^{\times} + Ce_{Ce,s}^{\times}$	Ce^{4+}/Ce^{3+} redox couple, carbonate-type adsorbate

Excessive carbon growth, resulting from undesirable CO disproportionation or deep reduction of CO_2, can lead to catalyst deactivation, microstructural damage, and performance loss under typical SOEC operating conditions.

- Coupled reaction mechanisms: RWGS and Non-Faradaic Electrochemical Modification of Catalytic Activity (NEMCA) effect

In the CO_2/H_2O co-electrolysis process of SOEC, the RWGS reaction is a significant influencing factor, rendering the mechanism of SOEC co-electrolysis more intricate. Some research indicates that CO_2 is predominantly transformed through RWGS due to the convergence of the polarization curves of H_2O electrolysis and co-electrolysis, along with comparable area-specific resistances (ASRs) (Fig. 9a) [45–47]. By contrast, other studies suggest that electrolysis and RWGS occur simultaneously during co-electrolysis. Ebbesen et al. [48], for example, reported polarization curves and impedance spectra for CO_2/H_2O co-electrolysis that differed clearly from those obtained for H_2O or CO_2 electrolysis alone, implying that both H_2O and CO_2 participate directly in the electrochemical process (Fig. 9b). Accordingly, the dominant pathway for CO formation remains controversial: CO may arise mainly from direct CO_2 electrolysis (R1) or from the RWGS reaction (R2). These

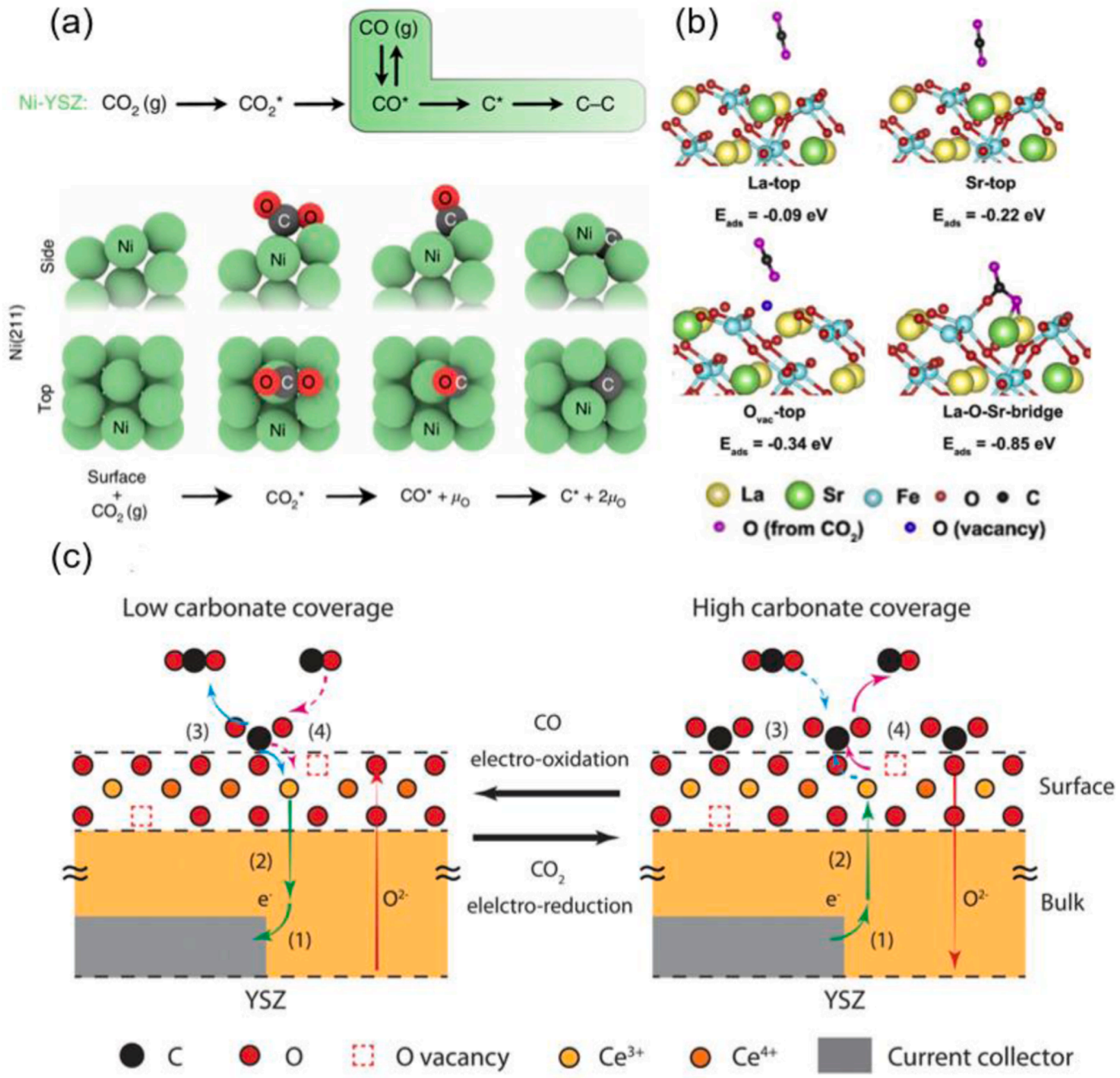

Fig. 8. (a) Proposed reaction mechanism on Ni and Reaction pathways for Ni-YSZ [41]. (b) The energetically most favorable CO_2 adsorption configurations on the $La_{0.75}Sr_{0.25}FeO_3$ (110) surface [42]. (c) The reaction pathways for SDC [43].

inconsistent conclusions are likely related to the strong sensitivity of the co-electrolysis mechanism to experimental conditions (Table 2).

As shown in Table 2, under typical SOEC co-electrolysis operating conditions (≥750 °C, Ni-YSZ cathode, H_2O/CO_2 feed ratio≈1), direct CO_2 electrolysis and the RWGS reaction occur simultaneously, with both contributing equally. Due to the system's high sensitivity to operating conditions and testing methods, various factors (temperature, configuration, gas composition) can alter the relative contributions of these two pathways; therefore, any unified attribution is unsound.

Besides operating parameters, the influence of the NEMCA effect on in-situ methanation via SOEC is also significant [56]. Under typical operating conditions, SOEC supply reaction ions (O^{2-} or H^+) via the Faraday process, where the reaction rate is generally proportional to the applied current or the ion transport rate. When a potential difference is applied, O^{2-} migrates from the solid electrolyte to the catalyst surface, resulting in the formation of an electrochemical double layer. This double electric layer modifies the electron density and chemical state of the catalyst surface, thereby affecting the adsorption and activation of reactants and influencing catalytic activity. Fujiwara et al. [52] conducted H_2O/CO_2 co-electrolysis in a button cell at 600 °C using a Ni-GDC cathode separately doped with Pd and Ru. The results showed that both the CO_2 conversion and methane selectivity increased with increasing applied potential, which can be explained by the NEMCA effect: polarization leads to an enrichment of electrons on the cathode surface, which may enhance its catalytic activity for CO_2 conversion via RWGS reaction (R2) (Fig. 10a). Another study demonstrated that its effect on the

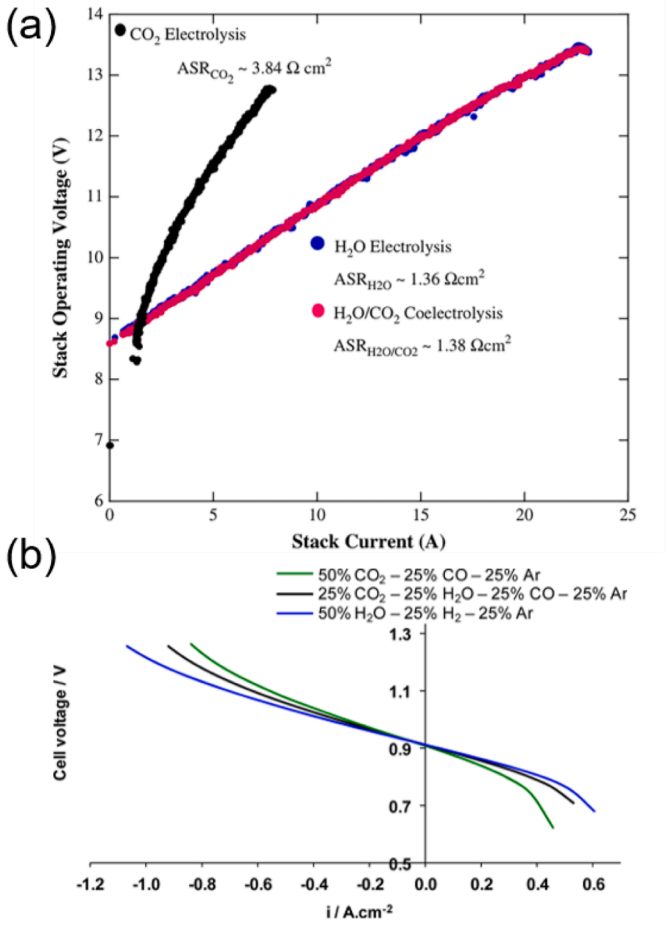

Fig. 9. (a) The polarization analysis of a 10-cell SOEC stack operated under comparable thermal conditions. Three feed scenarios were evaluated: steam electrolysis (54.8 % H_2O–22.5 % H_2–22.7 % N_2), co-electrolysis (54.9 % H_2O–22.6 % CO_2–22.5 % H_2), and pure CO_2 electrolysis (100 % CO_2) [49]. (b) Polarization curves for a planar Ni-YSZ|YSZ|LSM-YSZ SOEC under three distinct feeds: steam electrolysis (50 % H_2O–25 % H_2–25 % Ar), CO_2 electrolysis (50 % CO_2–25 % CO–25 % Ar), and co-electrolysis (25 % H_2O–25 % CO_2–25 % CO–25 % Ar) [48].

methanation reaction is also pronounced, and every rate-determining step (RDS) in the methanation sequence based on the SOEC was found to accelerate under the field [57], The study also indicated that the kinetic influence exerted by co-adsorbed oxygen atoms surpasses that of the electric field itself, implying that CO_2 methanation proceeds faster when the cell operates in SOEC mode, where oxygen coverage is lowered by electrochemical removal, as illustrated in Fig. 10b and c.

- Integrated reaction pathway for in-situ methanation

Based on the above discussion, a typical reaction network for in-situ methane synthesis via CO_2/H_2O co-electrolysis in SOECs can be summarized as comprising four key steps:

(1) Due to a lower reaction barrier of H_2O compared to CO_2, H_2O electrolysis tends to prevail initially during co-electrolysis.

(2) As more electrical energy is supplied, CO_2 electrolysis necessitates a higher reaction barrier for initiation, leading to simultaneous H_2O electrolysis alongside CO_2 through CO_2/H_2O co-electrolysis [55,58,59]. While a portion being reduced to CO at the electrode surface, another portion of the CO_2 that has dissociated and adsorbed onto the electrode surface undergoes hydrogenation to form CH_4 (methanation) or CO (RWGS).

(3) Hydrogenation of CO generated in situ (through co-electrolysis and RWGS) to form CH_4.

Table 2
Key factors influencing the reaction mechanisms of CO_2/H_2O co-electrolysis in SOECs.

Factor	Condition/ Parameter	Effect on the CO formation pathway	Key implication
Temperature [50]	600-700 °C	RWGS-dominated CO formation	Steam electrolysis prevails at a lower temperature
	>750 °C	Greater contribution from direct CO_2 electrolysis	Higher temperature enhances CO_2 activation and electrochemical kinetics
Cathode configuration [24]	Cathode-supported SOEC	Intermediate behavior between H_2O and CO_2 electrolysis	Cathode thickness affects the balance between electrochemical reactions and
	Electrolyte-supported SOEC	Similar to H_2O electrolysis	chemical reactions as a result of a mismatch between heterogeneous chemical reaction rates and electrochemical reaction rates
Electrode material	$Sr_2Fe_{1.5}Mo_{0.5}O_{6-\delta}$/ SDC [51]	Pathway-sensitive	Catalytic/transport properties govern reaction routes
	Ni/Pd–GDC; Ni-Ru-GDC [52]	Enhanced CO_2 conversion via RWGS	Polarization leads to an enrichment of electrons on the cathode surface, leading to high CO_2 conversion and methane selectivity
	LSCN-impregnated GDC-LSM oxygen electrode [53]	RWGS-dominated CO formation	During co-electrolysis, CO_2 conversion mainly occurs through RWGS, while H_2 is produced via H_2O electrolysis
Gas composition (H_2O/CO_2/ CO/H_2 ratio)	Increasing oxidant/ reductant ratio [27]	More direct CO_2 electrolysis	Higher oxidant ratios enlarge the electrochemical reaction region and align polarization behavior with CO_2 electrolysis
	H_2O concentration [54,55]	<5%: Similar to CO_2 electrolysis 5–30%: Mixed contribution >30%: RWGS and H_2O electrolysis dominate	The threshold concentrations of gas components
Voltage/ Current density [30]	High current density	Kinetically promotes CO formation	–
	Low voltage	Equilibrium-controlled behavior	

(4) Electro-reduction of water vapor produced during in-situ methanation prevents dissociation of methane through water vapor reforming [8].

Luo et al. [34] proposed the mechanism for in situ methanation in a two-temperature-zone SOEC:

(1) Cathode surface in the high-temperature region: acts as an inlet gas-control region where H_2 reacts with CO_2 to produce CH_4.

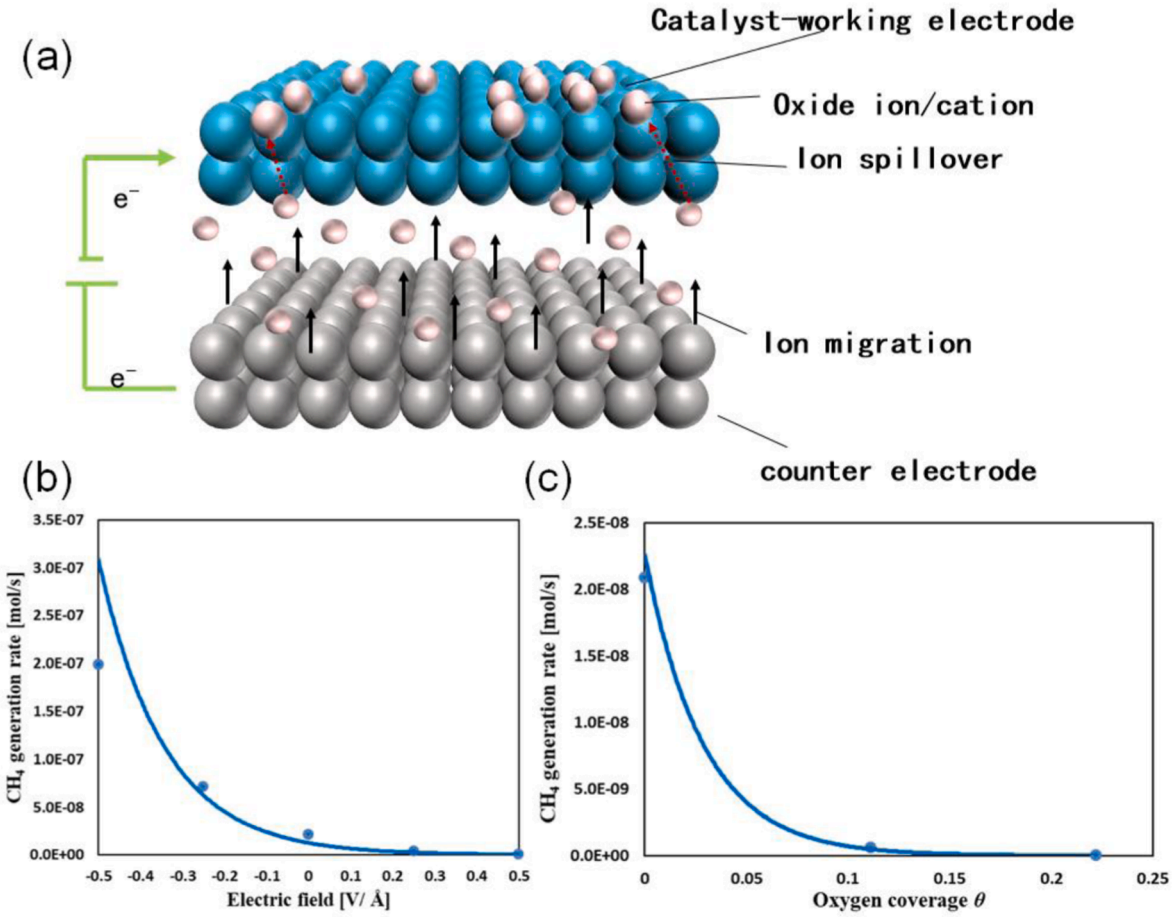

Fig. 10. (a) The principle of NEMCA. (b) Overall methane formation rate under externally applied electric fields of 0, ±0.25, and ±0.5 V $Å^{-1}$ [57]; (c) the influence of surface oxygen coverage (θ = 0, 1/9, and 2/9) [57].

(2) Positive electrode–electrolyte interface: serves as the electrochemical-control region where electrochemically produced H_2/CO rapidly form CH_4.
(3) Cathode surface in the low-temperature region: functions as a temperature-controlled zone where the methanation equilibrium shifts positively, facilitated by CO_2/CO methanation over a non-homogeneous catalyst.

In conclusion, there is still room for improvement of the intrinsic mechanisms of SOEC co-electrolysis and in-situ methanation. Mechanistic studies of CO_2/H_2O co-electrolysis reveal that while direct CO_2 reduction and RWGS are highly efficient at higher temperatures, methane formation is increasingly constrained by thermodynamics above 400 °C. This disconnect constitutes the root cause of the thermo-kinetic mismatch. Addressing this issue requires engineering solutions that optimize reaction pathways within a unified temperature range, which will be detailed in the following sections on material development strategies.

3. Design strategies for the materials of SOEC in-situ methanation

3.1. Cathodes (fuel electrodes): negotiating the thermo-kinetic mismatch in SOEC in-situ methanation

As discussed above, in-situ methanation in SOECs is not governed by a single elementary reaction, but by a strongly coupled network involving CO_2/H_2O co-electrolysis, RWGS, CO/CO_2 hydrogenation, and electrochemical promotion effects. The relative importance of these pathways varies significantly with the operating temperature, gas composition, current density, and local reaction environment near the cathode. In particular, the thermo-kinetic mismatch originates from an intrinsic conflict between electrolysis and methanation, and the key to bridging this mismatch lies in the development of multifunctional materials that can maintain sufficient electrochemical activity at intermediate or elevated temperatures while also enabling methanation to proceed within an expanded or shifted temperature window.

3.1.1. Traditional electrolytic optimization materials and their limitations

Traditional Ni-based cermets, including Ni-YSZ [60], Ni-SDC [61], and Ni-ScYSZ [62], have long dominated SOEC cathode design because of their excellent electrochemical activity for H_2O and CO_2 reduction. For example, Ni-YSZ electrodes with high Ni loading can achieve CO_2 conversion rates above 80%, with CH_4 selectivity approaching 99% under electrochemically assisted methanation conditions at 300–350 °C [63]. These results demonstrate that Ni-based electrodes can provide highly active sites for electrochemical reduction and, under suitable conditions, promote methane formation.

However, their application in direct in-situ methanation still faces intrinsic limitations. The surface of Ni-based cermets tends to favor the RWGS reaction, thereby promoting syngas production rather than selective methane formation, even under applied potentials [64]. Moreover, these electrodes usually require high operating temperatures above 700 °C to maintain sufficient ionic conductivity and electrolysis rates. In addition, Ni-based cermets are susceptible to carbon deposition, Ni coarsening, redox instability, and mechanical degradation in CO_2-rich or carbon-containing atmospheres, which raises serious concerns regarding long-term durability [65,66]. Although microstructural

engineering [67,68] and compositional regulation [69–72] have improved the electrochemical stability and durability of Ni-based cermets to some extent, these strategies mainly optimize conventional electrolysis performance and do not fundamentally resolve the temperature mismatch that limits direct methane synthesis.

In this context, mixed ionic-electronic conducting (MIEC) oxides have emerged as promising alternatives because of their improved stability in reducing atmospheres, tunable defect chemistry, and lower susceptibility to typical Ni-related degradation pathways such as carbon deposition and metal coarsening. Perovskite-based MIEC frameworks

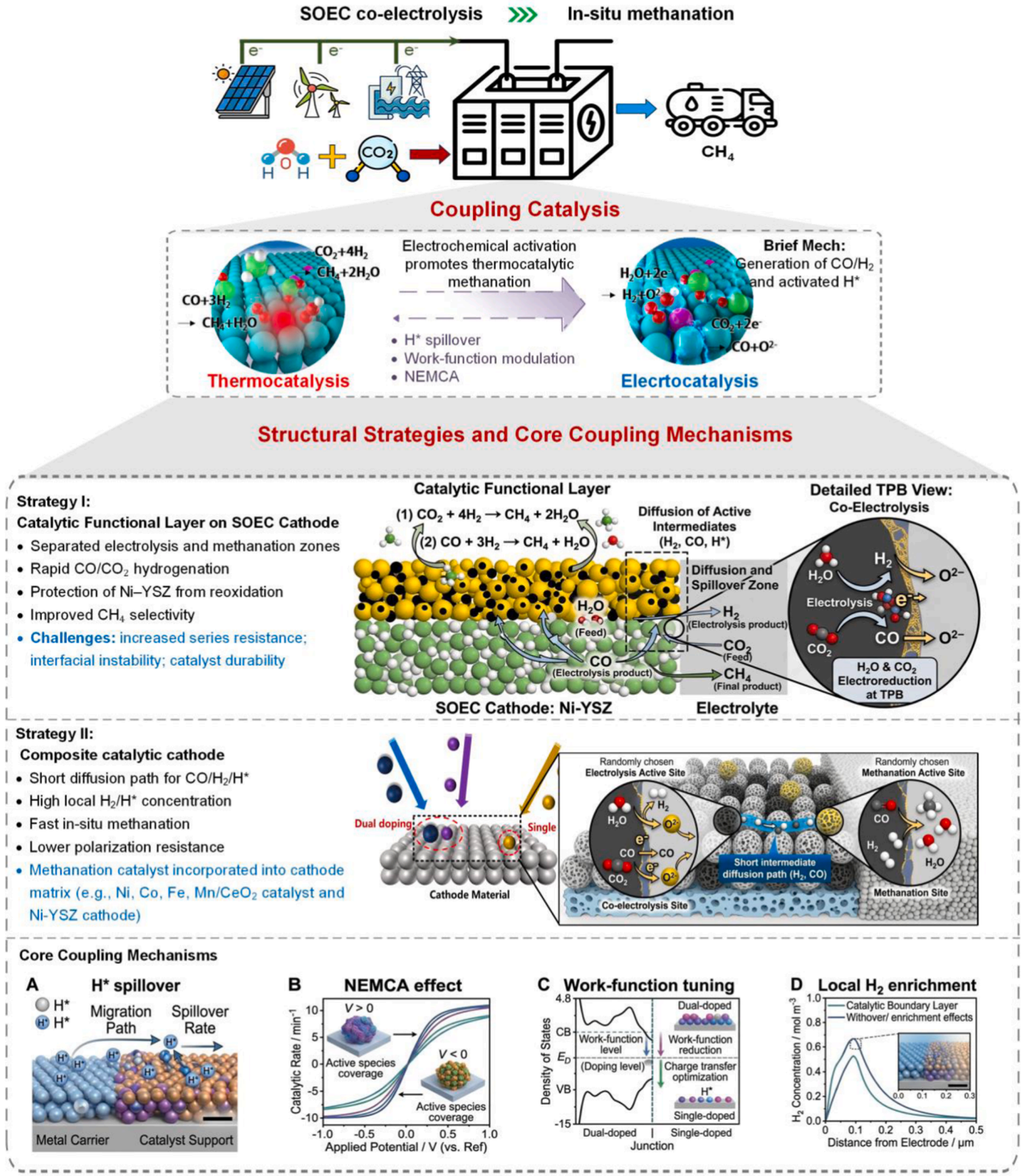

Fig. 11. Schematic illustration of thermo-electrocatalytic coupling in SOEC-based in-situ methanation.

[73–81] make use of adjustable oxygen-vacancy chemistry to enhance CO_2 adsorption, activation, and electroreduction kinetics, while enabling more flexible operation in medium-temperature ranges of approximately 600–700 °C.

Within these oxide frameworks, oxygen vacancies play a central role in binding and activating CO_2, while further performance optimization can be achieved through aliovalent doping [82–84], in-situ exsolution [85–87], and high-entropy modification strategies [88,89]. In particular, the incorporation of metallic nanoparticles such as Ni, Fe, and Co through B-site doping [90–92] or reduction-induced exsolution [93–95] has proven effective for constructing methanation-active sites on stable oxide backbones. For instance, recent studies on SFM decorated with exsolved Ni-Fe nanoparticles have shown that the perovskite matrix can anchor and stabilize active metallic sites, thereby enhancing hydrogenation kinetics and improving overall electrochemical performance [96–98].

Despite these advances, single-phase oxides and oxide-metal composite cathodes still generally fall short of delivering the high CH_4 selectivity and long-term stability required for practical SOEC-based in-situ methanation. This indicates that further material design must focus not only on improving electrochemical reduction activity, but also on strengthening the catalytic coupling between electrolysis-derived intermediates and methane-forming pathways.

3.1.2. *Multifunctional catalysts: coupling thermal catalysis and electrocatalysis*

The most direct response to the thermo-kinetic mismatch is to construct a multifunctional cathode that decouples or compensates for the opposing temperature requirements of electrolysis and methanation. Advances in electrode and electrolyte materials have enabled SOEC operation at progressively lower temperatures, which are often below 700 °C [99,100] and in some cases approaching 600 °C [101]. And partially alleviate the thermodynamic constraints on methane formation. Under these conditions, the synergistic interaction between electrochemical activation and catalytic hydrogenation becomes increasingly important.

Two principal approaches have been widely explored to realize thermo-electrocatalytic coupling (Fig. 11).

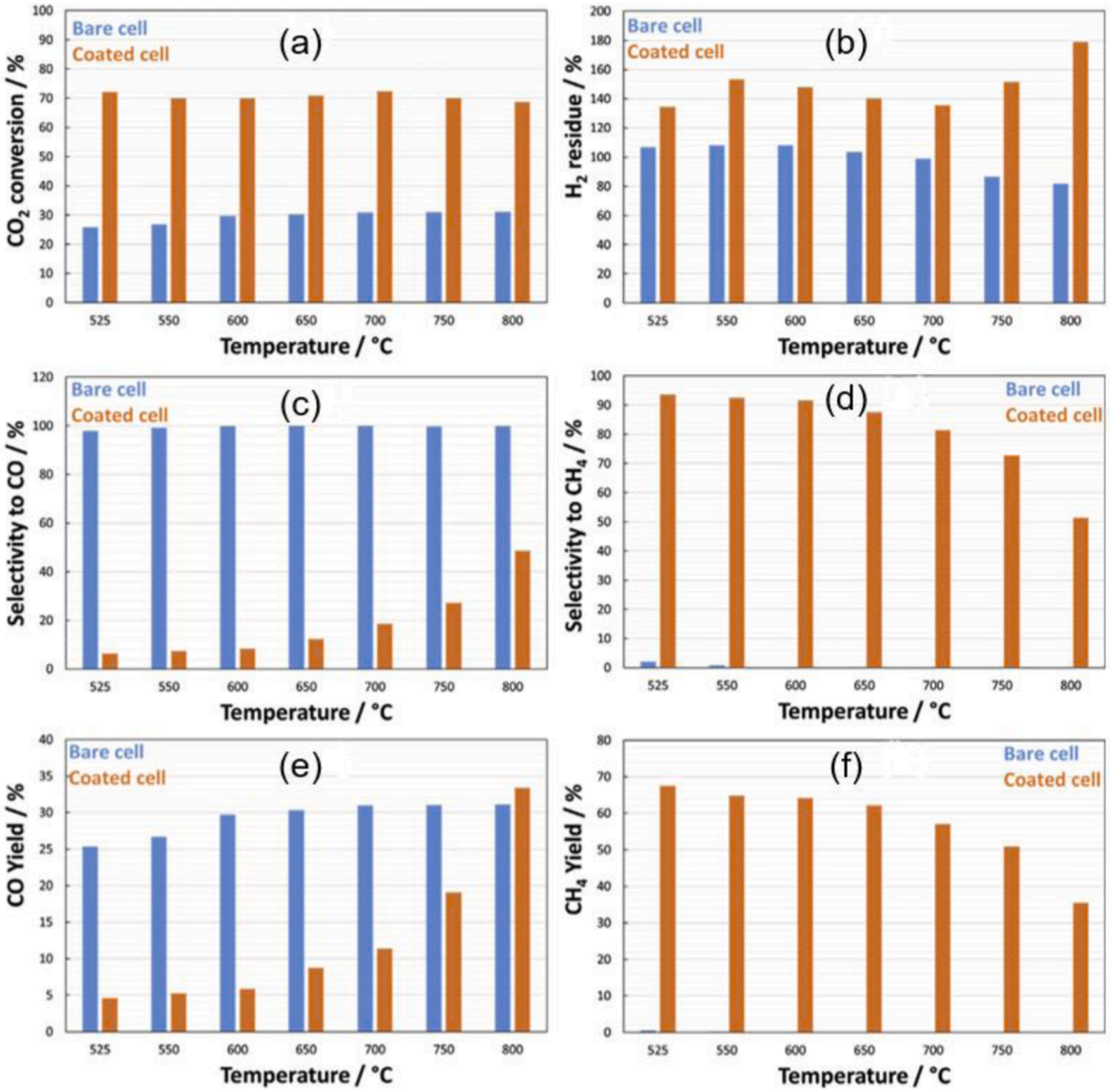

Fig. 12. The comparison of bare cell and coated cell (pre-layer Ni-Fe/CGO coating on the Ni-YSZ electrode): (a) CO_2 conversion; (b) H_2 residue; (c) Selectivity to CO; (d) Selectivity to CH_4; (e) CO yields; (f) CH_4 yields [104]: there is a significant enhancement in electrochemical CO_2 conversion, H_2 residue, CH_4 selectivity and direct methane production, surpassing the theoretical values for chemical conversion. Across nearly all temperature ranges, the bare cell exhibited a strong propensity to produce CO, with only minimal CH_4 generation at the lowest temperatures.

The first involves the introduction of catalytic functional layers on top of conventional SOEC cathodes. In this architecture, the underlying electrode performs electrochemical reduction of CO_2 and H_2O, generating CO and H_2 intermediates, while the overlying catalyst layer promotes rapid methanation. Functional layers containing catalysts such as CuSn-CGO, Ni-modified LSFCO ($La_{0.6}Sr_{0.4}Fe_{0.8}Co_{0.2}O_3$), or Ni-Fe/CGO [102–104] are typically deposited by impregnation, spraying, or screen-printing techniques. Representative studies demonstrate that the introduction of such catalytic coatings can significantly enhance methane selectivity, in some cases increasing CH_4 selectivity from negligible levels to above 90% in the temperature range of 525–600 °C [104] (Fig. 12). In addition to promoting methanation, the catalytic layer may also protect the Ni-YSZ cermet from reoxidation during operation.

However, the introduction of catalytic coatings can also increase the cell's series resistance and introduce mechanical instability at the cathode-electrolyte interface. Post-operation analyses frequently reveal microstructural fractures in this region, indicating that further optimization of catalyst distribution, porosity, and interfacial compatibility is required (Fig. 13). In addition, long-term catalyst durability remains insufficiently explored [105].

The second approach involves composite catalytic cathodes, where methanation catalysts are directly incorporated into the electrode structure. To optimize the utilization of medium-temperature SOEC under favorable operating conditions, it is imperative to introduce methanation catalysts that are highly selective and stable, facilitating seamless integration with the SOEC co-electrolysis process [106–109]. Among these, transition-metal catalysts supported on oxide matrices, which including Ni, Co, Fe, and Mn species dispersed on CeO_2-based supports, have shown promising methanation activity under electrochemically assisted conditions. And the Ni-based catalyst achieved a CO conversion of 98.5% at 375 °C, with CH_4 selectivity approaching 80% [110].

Bimetallic catalysts are especially attractive because their electronic and structural properties can be adjusted to achieve better catalytic performance and higher thermal stability. For example, adding precious metals such as Ru and Pd into Ni-GDC cathodes [56] can generate a hydrogen spillover effect from the noble metal sites to nearby Ni species, which improves CO methanation activity. Fe and Co are also commonly used because they are inexpensive and abundant [111]. As a result, Ni-Fe [71,112–114] and Ni-Co alloy [15] systems have been widely studied for their ability to tune syngas composition and enhance methane selectivity.

Mechanistic insights summarized in Section 2 provide an important basis for the design of thermo-electrocatalytic architectures. These studies show that an applied potential can alter catalyst surface chemistry and thereby enhance methanation activity, even under temperatures that would normally be unfavorable for CH_4 formation. A representative example is the Fe-Ag and Fe-GDC-Ag composite cathode reported by Biswas et al. [115], in which the increase in methane production was attributed to a synergistic interplay between higher current density (Fig. 14a and b) and potential-induced modification of the Fe work function. The change in the electronic state of Fe is believed to be crucial for promoting hydrogen spillover: atomic hydrogen (H*) generated by electrocatalytic H_2O dissociation can rapidly migrate through metallic pathways to neighboring methanation-active sites, thus directly coupling electrolysis with catalytic hydrogenation (Fig. 14c). Notably, although high temperatures generally disfavor methanation from a thermodynamic perspective, the combined effects of applied polarization, enhanced current density, and work-function regulation can shift the kinetic optimum toward higher temperatures, allowing maximum methanation performance to appear at around 600 °C. This case clearly illustrates that thermo-electrocatalytic coupling architectures are not simply physical combinations of electrolysis and catalytic components. Rather, they are material-level realizations of the NEMCA effect and hydrogen-spillover-assisted coupling mechanisms identified from mechanistic studies.

From this perspective, the NEMCA effect provides a route to partially overcome the thermodynamic and process constraints of conventional SOEC operation, opening a promising strategy for boosting methanation performance beyond what is expected from equilibrium considerations alone. To quantify the extent of electrochemical promotion in different systems, researchers have introduced normalized descriptors such as the rate enhancement ratio (ρ) and the NEMCA factor (|Λ|) which allow more direct comparison of electrochemical promotion effects across different catalyst and reactor configurations. The results in Table 3 confirm that electrochemical promotion is not limited to idealized model systems, but can also be realized under more realistic reactor conditions and comparatively high flow rates, and suggest that the magnitude of NEMCA promotion depends strongly on catalyst composition, the type of ionic conductor, and the applied polarization conditions. At the same time, evaluation of electrochemical promotion should not rely solely on ρ and |Λ|. Additional metrics, including the absolute enhancement of CH_4 formation rate, turnover frequency (TOF) changes under polarization, reduction in apparent activation energy, and Faradaic efficiency toward methane, should also be considered when benchmarking methanation performance. A more comprehensive assessment using these parameters will be important for clarifying the true catalytic contribution of electrochemical promotion and for guiding the rational design of next-generation thermo-electrocatalytic SOEC cathodes.

The rate enhancement ratio ρ is defined as:

$$\rho = \frac{r}{r_0}$$

r_0 denotes the open-circuit rate, while r represents the steady-state rate following polarization.

The NEMCA factor (epistemic Faraday efficiency) is defined as:

$$\Lambda = \frac{\Delta r}{I/nF}$$

Δr is the applied current, I is the applied current, and F is the Faraday constant.

NR = Not Reported.

However, quantitative evaluation of the NEMCA effect in SOEC-assisted methanation is still limited, experimental evidence regarding the identities of reaction intermediates and the catalytic pathways on polarized catalyst surfaces remains poorly understood, and the respective roles of electrocatalytic and thermal catalytic processes remain difficult to disentangle. Addressing these fundamental issues will be

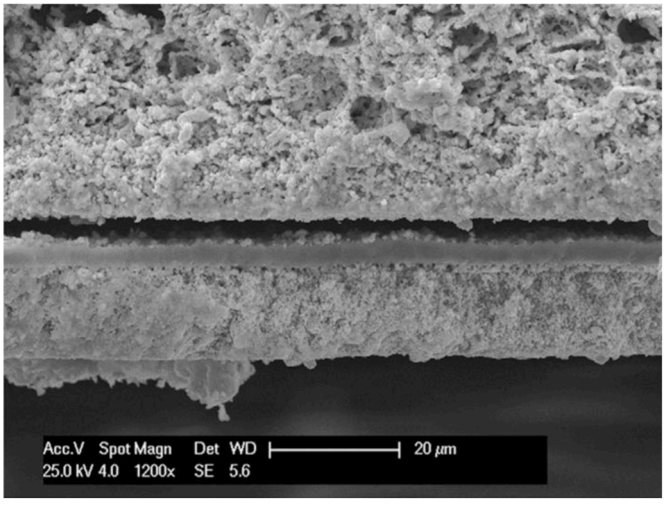

Fig. 13. Post-mortem SEM examination of the spent coated cell focused on the cathode–electrolyte interface, revealing microstructural features [104].

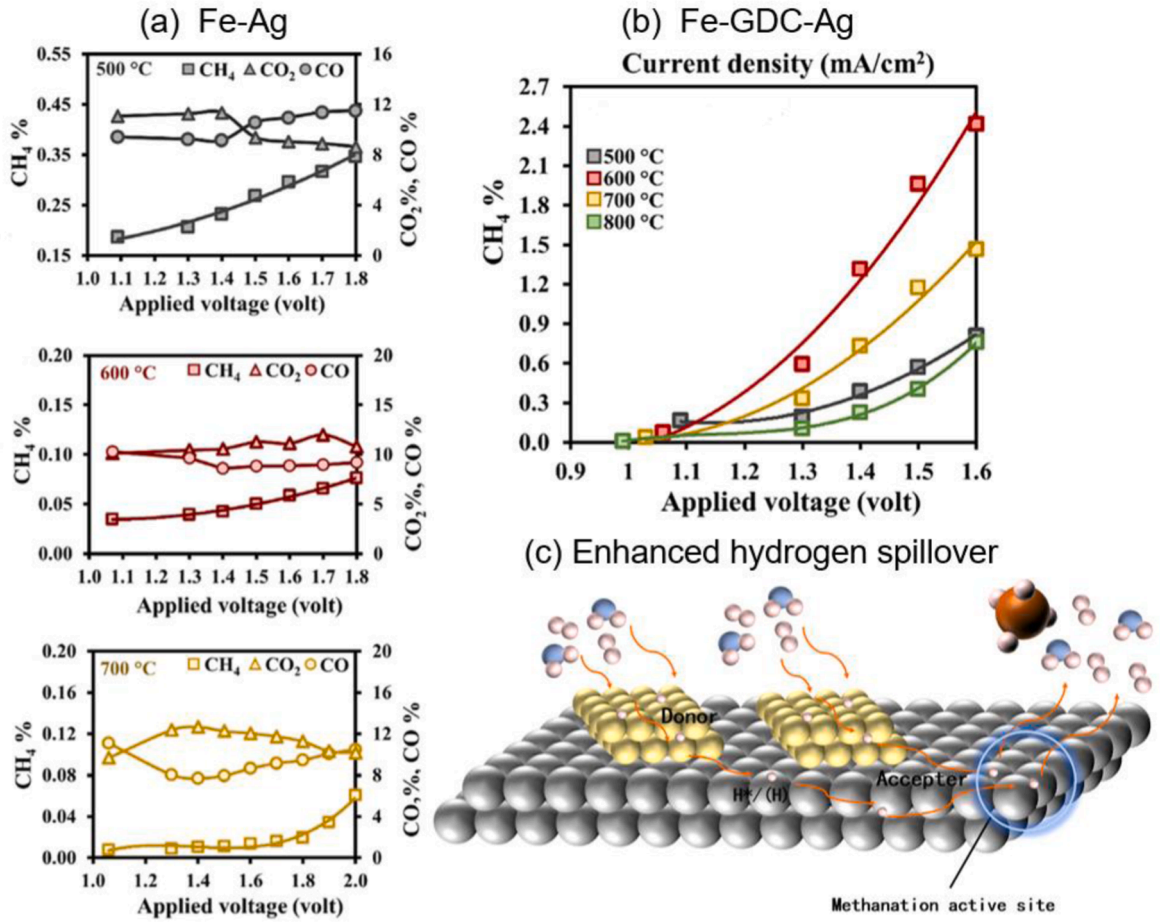

Fig. 14. Enhanced methanation performance of Fe-GDC-Ag relative to Fe-Ag under 1.0–1.8 V: (a) Fe-Ag electrode [116]; (b) Fe-GDC-Ag electrode [115], within the voltage range of 1.3–1.6 V, the Fe-GDC-Ag electrode exhibited nearly one order of magnitude higher methanation activity than Fe-Ag across all tested temperatures, accompanied by an approximately twofold increase in current density. The Fe-GDC-Ag system also delivered its highest methanation yield at 600 °C. (c) Proposed mechanism: on the Fe catalyst surface, H_2 (or hydrogen generated from electrolysis) undergoes dissociative adsorption, and the resulting H atoms spill over onto the GDC surface, where they participate in subsequent methanation steps.

essential for the rational design of next-generation thermo-electrocatalytic SOEC systems.

3.1.3. Representative strategies of cathode for SOEC-assisted in-situ methanation

Based on the above dialectical analysis, we could summarize three architectural strategies of cathode materials from the perspective of rebalancing the competition between electrolysis and methanation: (i) electrolysis-optimized cermets, (ii) catalytically tunable mixed-conducting oxides, and (iii) multifunctional architectures that directly couple thermal catalysis and electrocatalysis (Table 4). Conventional Ni-based electrodes generally favor syngas formation via the RWGS reaction, and their degradation (i.e., Ni coarsening) leads to a decline in the long-term stability of SOECs. The oxide-based electrodes mainly contribute to improved structural stability rather than strong methanation activity. By contrast, thermo-electrocatalytic coupling architectures have consistently delivered the highest methane selectivity, often exceeding 70–90%. These observations suggest that the most promising systems are those that integrate multiple functionalities through carefully engineered catalytic interfaces, which are most directly confront the temperature mismatch.

Future advances in direct methane synthesis in SOECs will likely rely on the rational integration of electrochemical intermediate generation, catalytic hydrogenation pathways, and thermo-electrocatalytic coupling mechanisms. Key research priorities include enhancing catalyst durability under long-term operating conditions, elucidating hydrogen spillover behavior and metal-support interactions, and quantitatively assessing electrochemical promotion effects.

3.2. Anode electrodes and electrolytes: matching the requirements of cathode-centered in-situ methanation

Compared with cathode materials, oxygen electrodes and electrolytes play a more indirect but equally important role in SOEC in-situ methanation. Their significance lies not only in improving electrochemical efficiency but also in determining whether the cathode can operate within a temperature range and interfacial environment suitable for the coupling of co-electrolysis and methanation. From this perspective, the design of oxygen electrodes and electrolytes should not be considered separately from cathode development. Instead, they should be evaluated according to how effectively they support cathode-centered medium-temperature operation, reduce ohmic and polarization losses, and maintain stable electrode-electrolyte interfaces under strongly coupled thermo-electrochemical conditions.

3.2.1. Anodes (oxygen electrodes): reducing polarization loss to mitigate the thermo-kinetic mismatch

Although methane is formed at the cathode, the oxygen electrode strongly influences the overall cell efficiency by determining polarization resistance, accessible operating temperature, and long-term stability. In particular, lowering oxygen-electrode polarization losses can reduce the high-temperature requirement typically needed for efficient

Table 3
Summary of NEMCA and electric-field promotion effects in CO_2 hydrogenation/methanation.

Reference	Catalyst	Electrolyte/Method	Temperature (°C)	Electric field/ Polarization	Rate enhancement ratio ρ	NEMCA factor |Λ|	Key kinetic effect
Kokkinou et al., 2023 [117]	Rh film	YSZ	380–430	±0.8 mA or ±1.5 V	$\rho(CH_4) = 1.7–11$; $\rho(CO) = 0.84–0.94$	$\|\Lambda(CH_4)\| \approx 0.23–0.62$; $\|\Lambda(CO)\| \approx 0.13–0.29$	Polarization enhances CH_4 formation and slightly suppresses CO; stronger CH_4 promotion under cathodic current; inverted-volcano dependence of CH_4 rate on potential
Kotsiras et al., 2018 [118]	Ru film/ Ru-Co catalyst	BZY	250–450	±1.5 V	$\rho(CH_4) \approx 1.3$; increases with catalyst loading	up to ~60	Anodic polarization promotes CH_4 formation and suppresses CO; cathodic polarization shows opposite behavior; CH_4 selectivity is tunable
Ruiz et al., 2024 [105]	Cu thin film	BZCY	325–550	−2 to+2 V	1.1–1.4	5.5–5.9	Moderate electrochemical promotion observed; optimal CH_4 selectivity (94.2%) at −0.5 V and 450 °C
Theleritis et al., 2012 [119]	Ru catalyst	YSZ	200–300	−1.3 to +1.3 V	~2–4	$10–10^3$	Oxygen ion supply from the electrolyte enhances CH_4 formation and suppresses CO formation
Wakamatsu et al., 2024 [57]	Ni (111), Ni (211), Ni (100)	DFT + microkinetic simulation	–	0–±0.5 V Å^{-1}	NR	NR	Electric field modifies adsorption energies and reaction barriers of rate-determining steps; CH_4 formation rate increases with field strength

Table 4
Representative strategies for SOEC-assisted in-situ methanation.

Design strategies	Representative system	T (°C)	Current density (mA cm^{-2})	CO_2 conversion (%)	CH_4 selectivity/yield (%)	Catalytic strategy
I. Electrolysis-dominated electrodes	Ni-YSZ [30]	800	470	66–75	41/11.8	Conventional Ni catalyst enabling electrolysis-driven methanation
	Ni-YSZ (high Ni loading) [63]	250–350	–	86	>99/~86	Highly dispersed Ni catalyst enabling low-temperature methanation
	Ni-GDC [56]	400	82	80	$CH_4/H_2 \approx 40\%/48\%$	NEMCA-induced promotion of CO_x methanation
	Ni-YSZ (doped Ni-SDC) [69]	700-800	2150-2960	enhanced	Limited CH_4 (Primarily SOEC electrolysis)	Composition-engineered Ni cermet favoring RWGS-driven syngas production
	Ni-Fe alloy [71]	~800	–	enhanced	limited CH_4	In-situ Ni-Fe alloy promoting RWGS and syngas formation
II. Catalytically tunable oxide electrodes	LST-CGO + Fe catalyst layer [26]	650	85	11.5	–/2.8	External Fe catalyst enabling thermocatalytic methanation
	NiFe@SFM-SDC [96]	800	2660	–	limited CH_4	Exsolved Ni-Fe nanoparticles with oxygen-vacancy enrichment
	LSTM-$Fe_{0.1}$ [120]	850	780	–	mainly CO	B-site Fe exsolution forming a stable metal-oxide interface
	$La_{0.5}Ba_{0.5}Mn_{0.7}Co_{0.3}O_{3-\delta}$ [121]	800–850	1010	–	mainly CO/H_2	Co-exsolution with redox-active perovskite surface
III. Thermo-electrocatalytic coupling architectures	Ni-Fe/CGO-coated Ni-YSZ [104]	525	150	72	**94/67**	Functional methanation coating layer
	Cu-Sn/CGO-coated Ni-YSZ [102]	525	150	30	**75/22**	Bimetallic catalytic layer for CO_2 hydrogenation
	Ni-LSCFO-coated Ni-YSZ [103]	525	150	4.1	7.6/1.1	Perovskite catalytic coating modifies surface kinetics
	Ru/Ni-GDC [52]	600	200	30	~45	Hydrogen spillover-enhanced methanation
	Pd/Ni-GDC [52]	600	200	18	~45	Electrochemical promotion of methanation (NEMCA)
	YSZ-SSC/GDC-coated Ni-YSZ [122]	450	14	44–70	**74–99.6**	Catalyst layer + downstream fixed-bed methanation
	Cu/BZCY/Ni-BZCY [105]	325–550	10–300	10–50	**50–80**	Electrochemically assisted methanation (strong NEMCA)
	Fe-Ag/Fe-GDC-Ag [115]	500–700	0.56–75	40–90	yield up to 2.42	Fe-mediated CO hydrogenation with GDC-assisted CO_2 reduction
	Co/Ni-YSZ [15]	500-700	120-150/–	57%	CH_4 concentration reached 2.1% at OCV	Bimodal Co distribution with high-surface-area active sites.

Methane selectivity and CO_2 conversion are reported when available in the literature; otherwise, representative electrolysis performance metrics are shown.

co-electrolysis, thereby enabling SOEC operation at lower temperatures where methanation is thermodynamically more favorable.

Early oxygen-electrode materials such as Sr-doped lanthanum manganite (LSM) [123–126] are not well suited for low-temperature SOEC operation because their predominantly electronic conductivity confines the oxygen evolution reaction (OER) to the three-phase boundary (TPB), resulting in a limited active area, sluggish kinetics, and a need for relatively high operating temperatures due to poor oxygen-ion transport. This further aggravates the mismatch between the optimal conditions for electrolysis and methanation. To address these limitations, mixed ionic-electronic conductors (MIECs), including $La_{0.6}Sr_{0.4}Co_{0.2}Fe_{0.8}O_3$ (LSCF) [127,128], $Ba_{0.5}Sr_{0.5}Co_{0.8}Fe_{0.2}O_3$ (BSCF) [129], $La_{2-x}Sr_xCo_{0.5}Ni_{0.5}O_{4\pm\delta}$ (LSCN) [130], and Co- and Sr-free alternatives like $Ba_{0.95}LaFeO_{3-\delta}$ (BLF) [131], have been widely explored, as their dual ionic/electronic conductivity extends the electrochemically active region beyond the TPB, thereby improving OER kinetics and lowering polarization resistance at medium temperatures (500–650 °C).

Moreover, oxygen electrodes with tailored microstructures [132], increased porosity, and improved surface oxygen exchange rates further enable efficient oxygen evolution at intermediate temperatures. This reduces the overall cell voltage needed for co-electrolysis and increases the fraction of thermal energy utilized, which is especially impactful in two-temperature-zone reactor designs. For example, the use of Ruddlesden-Popper (RP) oxides [133–136] and SFM-based double perovskites [137,138], has been shown to enhance durability and lower polarization resistance while preserving compatibility with cathode-side methanation conditions (Fig. 15). Furthermore, composite electrodes [139,140] and nanostructured electrodes prepared by impregnation, exsolution, or heterostructure construction [141], integrating complementary properties of multiple phases (e.g., electronic/ionic conductivity), exhibit superior electrical conductivity and electrochemical activity to synergistically support low-temperature cathode functionality.

Additionally, to support the development of proton-conducting electrolytes that are better suited for in-situ methanation, future research on oxygen electrode materials should focus on developing triconductive oxides (TCOs) capable of simultaneously conducting oxygen ions, electrons, and protons.

3.2.2. *Electrolytes: lowering the effective operating temperature and stabilizing cathode interfaces*

An ideal electrolyte for this application should have high ionic conductivity, negligible electronic leakage, sufficient gas tightness, and good chemical compatibility with adjacent electrodes. Just as importantly, it should maintain phase stability and strong interfacial adhesion under the wide oxygen partial pressure gradient encountered during SOEC operation [5,23,143–146]. For in-situ methanation, these requirements become even stricter because the electrolyte must support long-term operation under conditions where cathode materials are exposed to reactive CO_2, H_2O, CO, H_2, CH_4 mixtures and possible carbon-forming environments, which means the electrolyte should not be judged only by ionic conductivity under traditional SOEC conditions, but by whether it can help establish a lower-temperature operating window while maintaining chemical and mechanical stability.

Traditional YSZ-based electrolytes remain widely used because of their excellent chemical and structural stability; however, their relatively low ionic conductivity at intermediate temperatures generally requires SOECs to operate above 700 °C. Although such temperatures are acceptable for co-electrolysis, they are not ideal for methanation. In contrast, medium-temperature electrolytes such as Sc- or Sc/Y-stabilized ZrO_2, Ln_2O_3 (Ln = Dy, Er, Y, Gd, Nd, La)-doped δ-Bi_2O_3, Gd/Sm/Y-doped CeO_2, and perovskite oxides based on $BaCeO_{3-\delta}$, $BaZrO_{3-\delta}$, and $BaCe_xZr_{1-x}O_{3-\delta}$. Perovskite oxides, such as $La_{0.9}Sr_{0.1}Ga_{0.8}Mg_{0.2}O_3$ (LSGM) are more attractive for direct methane synthesis because they help lower the cell operating temperature. Among them, LSGM has shown particularly promising performance in methane-producing SOEC systems, with reports of 50% CH_4 yield at 550 °C and 1.3 V, and even higher CO_2 conversion under pressurized counter-current operation [8,17,104,122,143,147,148]. These results illustrate that electrolyte selection is closely tied to the ability of the cathode to function as both an electrolysis electrode and a methanation-active interface.

Proton-conducting perovskite electrolytes based on $BaZrO_3$, $BaCeO_3$, and their solid solutions have also received considerable attention because they can enable lower-temperature operation than oxygen-ion conductors [10]. Their lower activation energy for ionic transport makes them attractive for narrowing the thermo-kinetic mismatch. To improve both conductivity and stability, a large number of doped systems have been developed, including BZCYYb-type materials

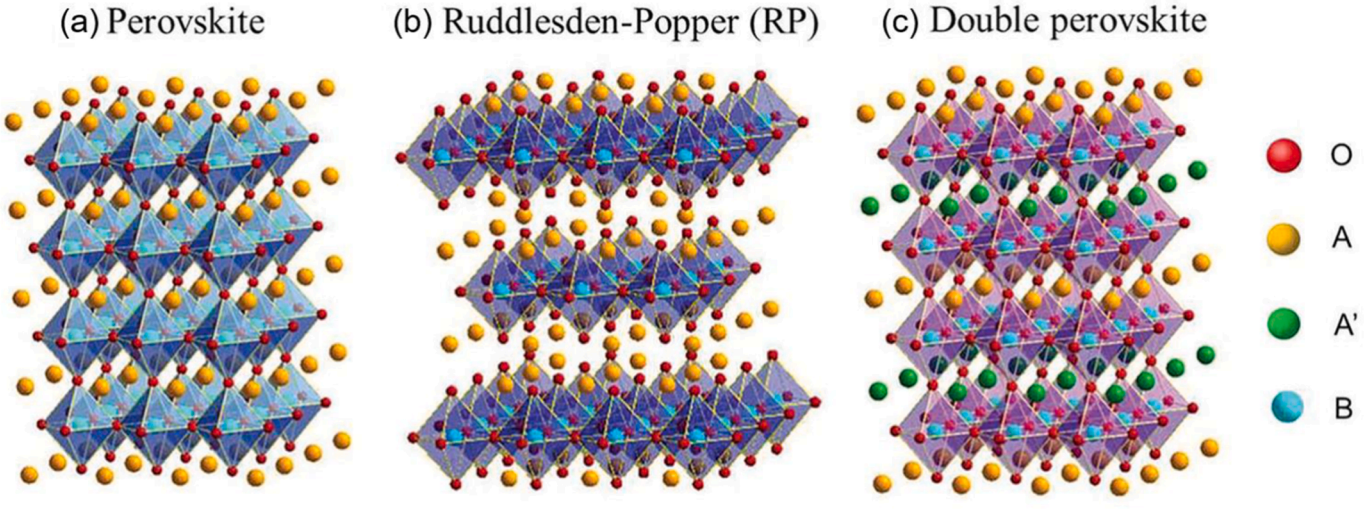

Fig. 15. (a) Crystal structures of perovskite structure oxides (ABO_3); (b) Ruddlesden–Popper structure oxides ($A_{n+1}B_nO_{3n+1}$): consist of alternating ABO_3 perovskite layers and AO rock-salt layers along the crystallographic c-axis, which can lead to a significantly lower formation energy for oxygen vacancies and interstitial oxygen species; (c) double perovskite structure oxide ($A_2BB'O_6$) [142]: B and B′ ions alternately occupy the centers of the octahedra, forming an ordered superlattice structure. This ordering allows for the control of the distribution and arrangement of oxygen vacancies, causing them to spontaneously form one-dimensional or two-dimensional ordered channels.

[149–151]. Chen et al. [152] further reported Si-doped BZCYYb, which retained good electrochemical performance while improving mechanical hardness. From the perspective of direct methanation, the importance of these proton-conducting electrolytes lies in their potential to shift the entire cell toward a more methanation-compatible temperature regime. In addition to the development of novel low- and medium-temperature electrolyte materials, nanotechnology, high-entropy modification, and other material enhancement techniques have demonstrated significant potential in enhancing the ionic conductivity and durability of low-temperature SOEC electrolytes. This improvement can be attributed to their unique surface and interfacial properties, as well as the effects associated with nano- and high-entropy materials [152,153].

At the same time, electrolyte compatibility with the cathode cannot be overlooked. Some promising electrolytes, such as LSGM, may react with conventional Ni-based fuel electrodes during fabrication or long-term operation [154]. This means that a high-conductivity electrolyte is not necessarily an optimal choice unless it can be matched with cathode materials that maintain interfacial stability. Therefore, electrolyte development for in-situ methanation should follow a co-design principle, in which ionic transport properties, sintering behavior, chemical compatibility, and cathode integration are considered together.

Beyond compositional optimization, reducing electrolyte thickness is another effective approach for decreasing ohmic resistance and expanding the medium-temperature operating window. Accordingly, electrode-supported thin-electrolyte configurations have become increasingly important in direct-methanation SOECs.

More broadly, oxygen electrodes and electrolytes should be regarded as supporting components within a cathode-centered in-situ methanation system, rather than as isolated elements optimized independently. Future material development should move beyond the separate optimization of individual cell components and instead focus on the integrated matching of cathode, oxygen electrode, and electrolyte. In the context of direct methane synthesis, the goal is not simply to maximize electrolysis current density, but to establish a stable and efficient operating window in which electrolysis and methanation can proceed synergistically. From this perspective, oxygen electrodes should minimize the kinetic penalties associated with lower-temperature electrolysis, while electrolytes should reduce ohmic losses and maintain stable interfaces with multifunctional cathodes. Only through this kind of system-level materials coordination can the full potential of SOEC-based in-situ methanation be realized.

4. Reactor design for SOEC in-situ methanation: addressing thermo-kinetic mismatch

The mismatch of temperature makes reactor design more than a simple issue of cell geometry. Its main role is to create a reaction environment in which electrolysis and methanation can be coupled as effectively as possible [20,25]. From this viewpoint, current reactor designs can be discussed according to how they deal with the temperature mismatch and organize the reaction field.

4.1. Configuration comparison: single-temperature-zone vs. two-temperature-zone reactors

4.1.1. Single-temperature-zone reactors

Single-temperature-zone systems are the simplest approach to reactor design, where the cathode acts as both the electrolysis electrode and the methanation-active surface, or is directly coupled with an additional catalyst layer. Planar SOECs are the most common implementation (Fig. 16a), offering structural simplicity and high power-density. Fuel-electrode-supported cells reduce ohmic losses through thinner electrolytes, making lower-temperature operation feasible and slightly favoring methanation. Metal-supported cells may provide additional benefits in heat management, although their long-term durability still requires further validation [155]. However, methane formation remains thermodynamically and kinetically limited under uniform high-temperature conditions, as demonstrated by low yields from electrolyte-supported (~0.2% CH_4 [26]) and cathode-supported (~0.286% CH_4 [27]) SOECs. Although increased pressure and gas flow optimization can promote methane formation, single-zone designs fundamentally cannot resolve the intrinsic temperature mismatch between electrolysis and methanation, and their practical performance continues to rely heavily on improvements in cathode materials and catalyst design which discussed in section 3.1.

4.1.2. Two-temperature-zone reactors

In two-temperature-zone configurations, electrolysis and

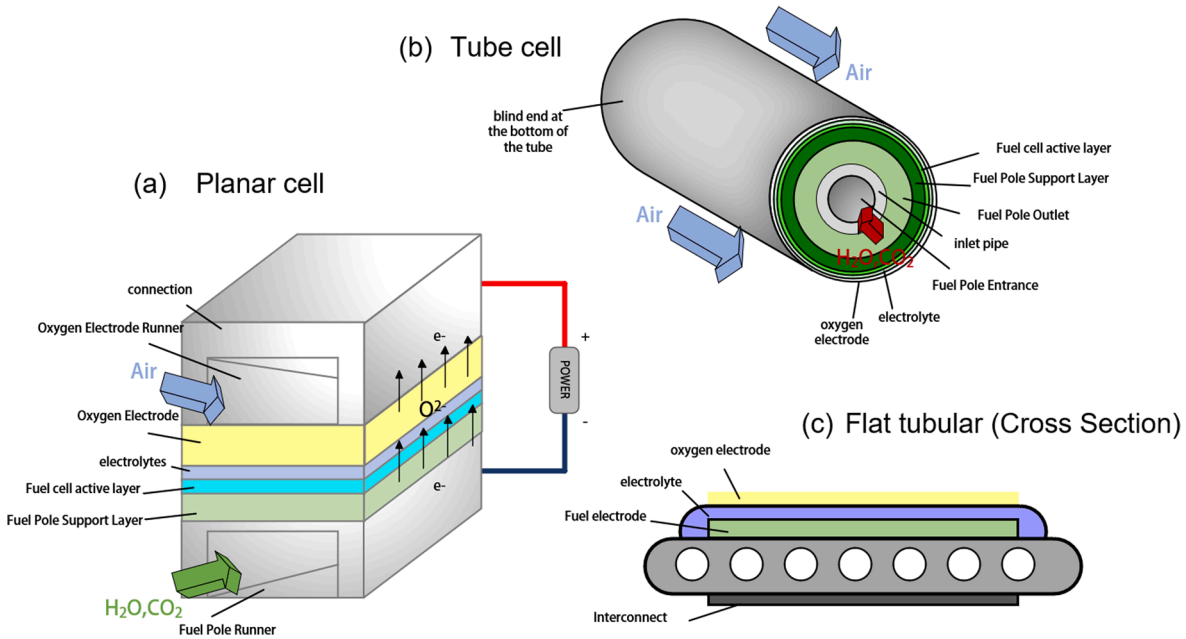

Fig. 16. Typical (a) planar cell; (b) tube cell; (c) flat tubular cell.

methanation are spatially separated so that each process can proceed under its respective optimal conditions. Tubular and microtubular reactors are particularly well suited to this strategy (Fig. 16b), as their axial geometry naturally enables temperature zoning. For example, Chen et al. [30] reported a CH_4 yield of 11.84% in a tubular reactor, while Lei et al. [29] further increased the yield to 21.1% using a microtubular design. Additional improvements in CO_2 conversion and methane yield can be achieved through pressurization [16] and optimization of the relative lengths of the two temperature zones [156].

Despite their advantages, dual-zone reactors pose considerable engineering challenges, especially in elongated tubular systems, where axial temperature gradients can generate substantial thermomechanical stress and thus compromise structural integrity [155]. Therefore, maintaining stable temperature gradients and alleviating thermal stress are key considerations in the design of two-temperature-zone SOEC reactors.

Flow-field optimization has emerged as an effective strategy to address these issues. Reported studies indicate that rotating L-shaped flow fields can achieve the most uniform gas distribution and the highest hydrogen production efficiency, although thermal stress may become concentrated at rib edges. In contrast, parallel flow fields exhibit the lowest thermal stress and are therefore more suitable for durability-oriented designs [157]. Rectangular flow channels have also been shown to improve fuel utilization compared with square and trapezoidal geometries, albeit at the expense of less uniform stress distribution [158]. Meanwhile, cross-flow configurations provide a promising balance between electrochemical performance and mechanical stability by simultaneously enhancing hydrogen production efficiency and reducing thermal stress [159]. Counter-flow arrangements further improve temperature uniformity, while the use of metal foam as a flow distributor can lower both the average temperature and thermal stress, although this may involve trade-offs in pressure drop and fabrication complexity [160]. Beyond flow-field design, additional thermal-management strategies have also proven effective in improving temperature uniformity and relieving stress throughout reactor components. These include external air cooling [161], the integration of planar high-temperature heat pipes [162], regulation of inlet gas temperature, and dynamic adjustment of anode gas flow [163]. Collectively, these approaches not only enhance SOEC performance and durability, but also provide important guidance for the design and long-term stable operation of dual-zone tubular reactors.

To combine the respective advantages of planar and tubular architectures, flat-tubular SOECs have also been proposed [35] (Fig. 16c). These systems offer a potential compromise between compactness and mechanical robustness. However, research on flat-tubular SOEC reactors remains relatively limited. Further experimental and modeling studies are still needed to optimize operating conditions, understand thermal matching among materials, and establish the theoretical basis required for structural design and practical engineering implementation of flat-tubular SOECs [164]. Several common reactor configurations for direct methanation by SOEC are shown in Table 5.

Table 5
Comparison of reactor Configurations for SOEC in-situ methanation.

Reactor Configuration	Thermal Management Complexity	Mechanical Stress	CH_4 Yield (%)	Scale-up Potential
Planar (single-zone)	Low	Moderate	<1	High
Tubular (dual-zone)	High	High	10–21	Moderate (fabrication limits)
Flat-tubular	Moderate	Moderate	Limited data (<5)	Promising, but unproven
Microtubular (dual-zone)	High	High	Up to 21	Experimental scale only

4.2. *Flow-field designs: the potential to mitigate the temperature mismatch*

The role of flow-field engineering in thermal management extends beyond the mitigation of thermal stress, as it also offers an effective route to alleviate thermo-kinetic mismatch. Advanced flow-channel geometries, including trapezoidal [165], elliptical [166], and X-type designs [167], can improve localized heat transfer by modifying gas-flow behavior and enhancing interactions between reactants and reactive surfaces. In doing so, these configurations promote more uniform heat and mass transfer, reduce local "dead zones," and facilitate more consistent oxygen transport to active reaction sites. For example, staggered or discrete rib structures [168] enable multidirectional gas diffusion while reducing oxygen concentration gradients beneath the ribs, thereby shortening diffusion pathways and accelerating reactant replenishment. These effects are particularly relevant to SOECs, where improved transport and thermal uniformity can help relieve local temperature–reaction mismatches.

More sophisticated geometries, such as trapezoidal flow channels [165], offer additional advantages by extending gas residence time. In SOEC systems targeting methane production, longer residence times for CO_2, H_2, and CO increase contact between gaseous reactants and catalytic sites, thereby promoting the hydrogenation of CO to methane. At the same time, the gradual tapering of ribs and channels helps optimize flow distribution, suppress hot-spot formation, and maintain more uniform heat transfer along the channel. Such features are especially important in two-temperature-zone SOECs, where precise regulation of reactant transport and temperature is required to balance the distinct demands of electrolysis and methanation.

Although most of these advances in flow-field design have been developed in the context of SOFCs, the underlying principles are highly transferable to SOEC systems. Therefore, adapting optimized flow fields and current-path designs from SOFC research holds considerable promise for improving SOEC operational stability and methane-production efficiency. In this sense, flow-field engineering provides not only a practical foundation for enhancing in-situ methane synthesis, but also a broader design framework for SOEC applications requiring precise thermal and kinetic control.

In addition, appropriate catalyst placement within the flow field is critical for thermally intensified reactor designs. Direct catalyst coating on the cathode surface (Fig. 17a) can enhance methane selectivity, with reported CH_4 selectivity reaching 44%, but this strategy also increases sensitivity to temperature fluctuations. By contrast, combining cathode-supported catalyst coating with a downstream fixed-bed catalyst [122] can significantly improve both CO_2 conversion (~70.5%) and CH_4 selectivity (~99.6%) by extending gas residence time and redistributing the heat released during methanation (Fig. 17b and c).

Overall, single-temperature-zone reactors are simple, but their methane yield is limited by the narrow overlap between electrolysis and methanation conditions. Two-temperature-zone reactors provide a more direct solution by separating the two functions in space, although they also introduce greater thermal and mechanical complexity. Flow-engineered and thermally intensified reactors further improve local matching by optimizing heat transfer, residence time, and catalyst utilization. To mitigate the temperature mismatch, high-performance reactor design for SOEC in-situ methanation should be understood as a combined problem of geometry, support configuration, catalyst placement, pressure, and flow arrangement [17,18,32,34,169,170], rather than a simple choice of reactor shape.

5. Conclusion and outlook

The development of SOEC-based in-situ methanation is fundamentally limited by the thermo-kinetic mismatch between high-temperature co-electrolysis and low-temperature methanation. This mismatch represents the core scientific and engineering challenge of the field and also

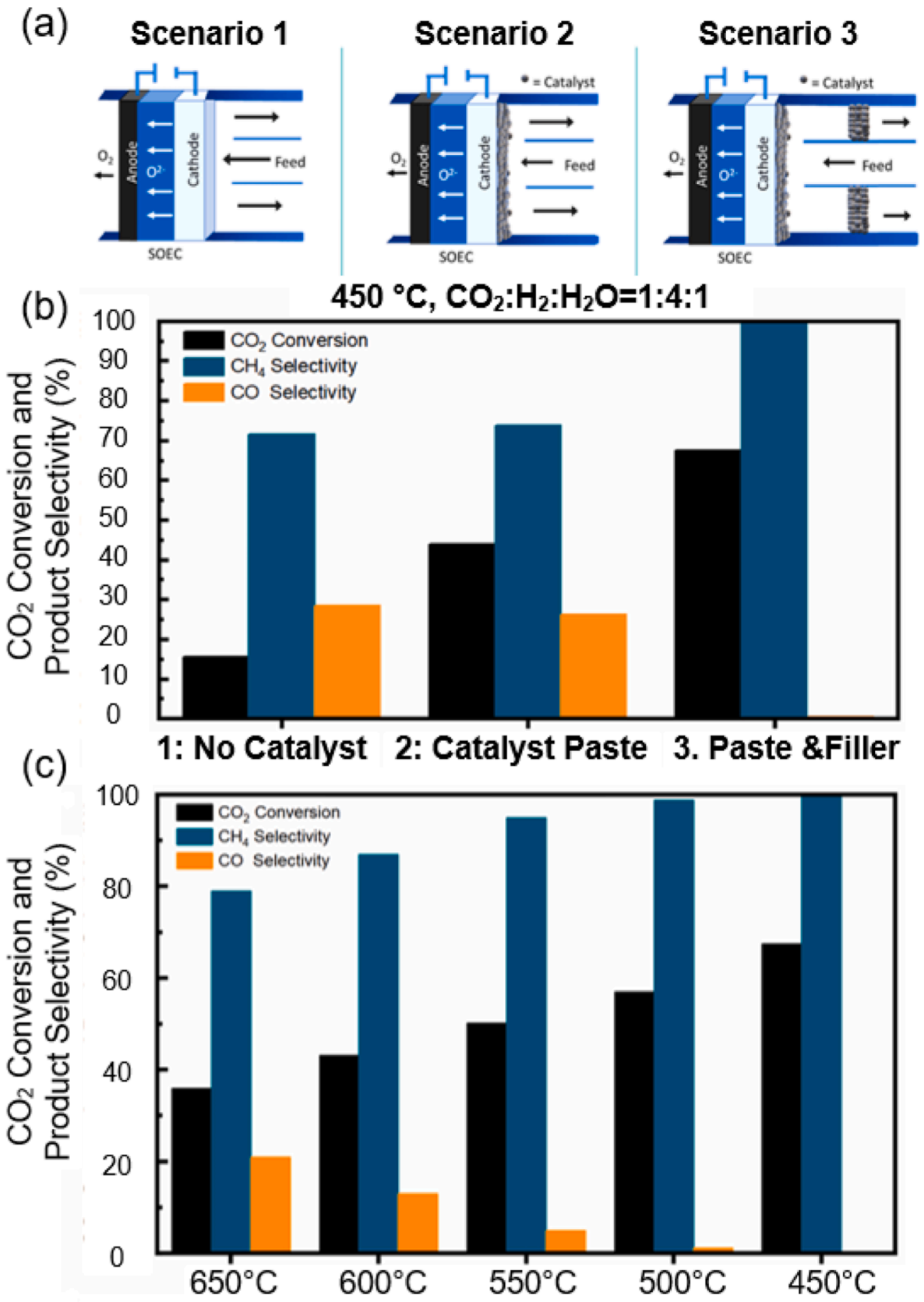

Fig. 17. (a) 3 Case scenarios for catalyst implementation: 1) No catalyst, 2) A catalyst paste over the cathode, and 3) Catalyst over the cathode and in the effluent line as filler (fixed bed); (b) methanation data for 3 case scenarios of methanation catalyst implementation over a SOEC; (c) OCV methanation data for the scenario 3.

explains why meaningful progress cannot rely on the isolated optimization of any single component, but instead requires the coordinated advancement of reaction mechanisms, functional materials, and reactor configurations.

From a mechanistic standpoint, methane formation in SOECs results from the coupling of several pathways, including direct electrochemical CO_2 reduction, the RWGS reaction, thermocatalytic hydrogenation, and carbon-related side reactions. Nevertheless, the relative contributions of these pathways under realistic operating conditions remain insufficiently understood. Although electrochemical promotion effects, such as NEMCA and hydrogen spillover, are increasingly considered to be potentially important in facilitating methane formation, their quantitative roles under practical SOEC conditions have not yet been clearly established.

At the materials level, current research is moving beyond conventional electrolysis-oriented cathodes toward multifunctional cathode architectures that combine co-electrolysis activity with methanation capability. Oxygen electrodes and electrolytes should be selected and designed according to how effectively they support cathode-centered operation in the medium-temperature range, reduce polarization and ohmic losses, and maintain stable interfaces with methanation-active cathodes. The goal of materials design is not simply to maximize electrolysis current density, but to establish a stable and efficient operating window in which electrolysis and methanation can proceed in a coordinated manner.

At the reactor level, current research clearly shows that reactor engineering is an important means of mitigating the thermo-kinetic mismatch. Single-temperature-zone reactors are structurally simple, but their methane productivity is constrained by the limited overlap between electrolysis and methanation conditions. Two-temperature-zone reactors offer a more direct solution by spatially separating the high-temperature electrolysis region from the lower-temperature methanation region, while thermally intensified and flow-engineered designs further improve local compatibility through control of heat transfer, gas residence time, catalyst distribution, and pressure. These studies indicate that efficient methane production depends not only on reactor geometry, but also on reactor-operation coupling, including the coordinated optimization of temperature, pressure, flow mode, and catalyst placement.

At the system level, one of the principal advantages of in-situ methanation lies in its intra-system heat utilization, namely, the direct feedback of methanation heat to support the endothermic electrolysis process [8]. In this review, the temperature mismatch issue associated with SOEC-based in-situ methane production has been systematically analyzed, and feasible strategies for temperature zoning have been discussed in detail. Beyond reactor-level thermal matching, however, efficient heat integration at the system level is equally critical to the engineering viability of the overall SOEC-to-methane process. A first key approach is heat-exchange design. By integrating heat exchangers between the methanation and electrolysis zones, the exothermic heat released during methanation can be transferred to preheat the electrolysis feed, thereby reducing the external energy input. A representative example is the design in which a copper tube was inserted at the SOEC outlet to introduce low-temperature CO_2, allowing it to mix with the high-temperature cathode off-gas and thus realize direct thermal recovery [171]. A second important strategy is the incorporation of thermal storage units at critical locations, such as thermal energy storage (TES) [172] modules or liquid CO_2 energy storage tanks (LCES) [173]. These components can buffer transient thermal loads, suppress temperature fluctuations, and improve operational stability under dynamic conditions. At a broader process scale, coupling the SOEC-methanation system with thermodynamic power cycles offers an effective route for valorizing surplus heat [174]. For example, integration with an organic Rankine cycle (ORC) has been shown to enable substantial heat recovery in O-SOEC-based CO_2-to-CH_4systems, leading to a methane production efficiency of 86% and an overall process efficiency of 77% [175]. In another case, combining an ORC with a double-effect Li/Br absorption refrigeration cycle (ARC) in a geothermal-power-driven CO_2-to-CH_4 system reduced the energy demand for methane production by 35% and shortened the payback period by 19%. Notably, the use of isopropanol as the ORC working fluid further reduced the payback period to 7.419 years [176].

Additionally, Long-term stability is still one of the core bottlenecks in the engineering application of SOEC in-situ methanation. Compared with conventional SOEC, the SOEC-methanation coupling system simultaneously withstands high-temperature electrolysis, strong reducing atmosphere, exothermic methanation reaction and complex atmosphere composition changes, and its degradation mechanism is more diverse and coupled. Firstly, the Ni-based fuel electrode/catalyst phase is prone to particle coarsening, migration and redistribution under high temperature, high water vapor partial pressure or strong local temperature gradient, resulting in a decrease in the length of the three-phase interface, a decrease in electrochemical activity, and a possible weakening of methanation catalytic activity. Secondly, due to the significant thermodynamic and kinetic differences between the electrolysis zone and the methanation zone, the local thermal expansion mismatch may induce electrode-electrolyte interface peeling, contact resistance increase and mechanical crack propagation, especially in the process of start-stop, load fluctuation and temperature zone switching. In addition, under the co-existence atmosphere of CO_2/CO/H_2, side reactions such as Boudouard reaction, CO disproportionation and methane cracking may lead to carbon deposition, blocking channels, covering active sites and accelerating the destruction of electrode structure. The above process not only reduces the electrolysis performance and methane generation efficiency, but also further enlarges the local temperature and concentration gradient, forming a positive feedback instability of 'heat-mass transfer-structure degradation'.

Despite these promising advances, the commercialization of direct methane synthesis via SOEC co-electrolysis still faces substantial challenges. Future research should therefore focus on progress in the following areas:

(1) In-Depth Development of Catalysts and Cell Materials: Future efforts should focus on designing cathode, anode, and electrolyte materials that operate efficiently within the medium-temperature range (~550 °C) with high ionic-electronic conductivity, superior methanation activity, and strong resistance to carbon deposition, and enhancing the compatibility between electrolytes and catalytic cathodes under low-temperature operating conditions. Greater emphasis should be placed on understanding and exploiting the synergistic interactions between bimetallic or multimetallic alloy nanocatalysts and cathode substrates, including cermet and perovskite-based frameworks. Approaches such as high-entropy engineering [177] and elemental doping offer effective means to regulate the intrinsic electronic structure of materials as well as their oxygen-vacancy concentration. In parallel, techniques such as infiltration and in situ exsolution should be employed to construct nanostructured active sites with extended triple-phase boundaries and strengthened metal-support interactions. Furthermore, the integration of artificial intelligence with high-throughput computational methods is expected to accelerate the discovery, screening, and performance prediction of emerging material systems, including high-entropy oxides and ternary mixed ionic-electronic conductors.

(2) In-situ Resolution of Reaction Mechanisms and Interfacial Processes: The quantitative contributions of individual reaction pathways remain insufficiently clarified, particularly with respect to the NEMCA effect and its distinction from other polarization-induced phenomena, such as local H_2 enrichment, oxygen-vacancy formation, work-function modulation, and interfacial charge transfer. At the same time, there is an urgent

need to develop and apply advanced in-situ and operando characterization techniques, including APXPS and operando Raman/SEM, to track in real time the evolution of active sites, the formation of reaction intermediates, and the pathways of ionic and electronic transport at the triple-phase boundary under realistic operating conditions. Such efforts will provide direct mechanistic insight and a stronger theoretical basis for material design and operating-parameter optimization.

(3) System Integration and Long-Term Stability Enhancement: Future research should place greater emphasis on clarifying the fundamental causes of degradation, including nickel migration and agglomeration, electrode poisoning, and sealant failure. An integrated strategy that combines theoretical simulation with in-situ and operando characterization is essential for revealing both reaction and degradation mechanisms at the atomic scale. Density functional theory can be used to predict reaction pathways and associated energy barriers, while techniques such as in-situ X-ray diffraction, X-ray photoelectron spectroscopy, and environmental transmission electron microscopy provide real-time insight into the structural and chemical evolution of electrode materials. Beyond reporting short-term electrochemical performance and methane yield, future studies should also establish a systematic framework for long-term durability assessment relevant to engineering applications, including constant-current and dynamic-load testing, thermal cycling, start-stop cycling, carbon-deposition tolerance evaluation, and post-failure multiscale structural characterization. The combination of degradation mechanism analysis and reactor design optimization can provide more reliable design basis for life prediction, material screening and engineering amplification of SOEC in-situ methanation system.

Overall, SOEC in-situ methanation is evolving from proof-of-concept studies toward a more integrated scientific and engineering framework. At present, two bottlenecks appear to be particularly decisive for future progress. One is the absence of a quantitative mechanistic description of NEMCA-type electrochemical promotion under realistic methanation conditions, which limits the rational design of thermo-electrocatalytic cathodes. The other is the insufficient compatibility of existing electrolyte/electrode systems with the medium-to low-temperature regime required for selective methane formation, where ionic transport, interfacial stability, and long-term durability become increasingly critical. Future research should therefore prioritize, first, operando and modeling-based quantification of electric-field effects, surface oxygen coverage, and competitive methane-formation pathways; and second, the development of cathode-centered integrated materials systems that can sustain efficient co-electrolysis and methanation under medium-temperature conditions. Combined with reactor designs that enable precise thermal and flow-field regulation, these advances will be essential for translating SOEC-based direct methane synthesis into a practical power-to-methane technology.

Author contributions

Feiyang Liao: Writing – original draft, Investigation, Conceptualization; **Junbo Xu**: Resources, Writing – review & editing; **Chao Yang**: Funding, Supervision, Review; **Wei Zhou**: Validation, Resources, Methodology.

Funding

This work was funded by the National Natural Science Foundation of China (22421003, 22378393, 22508388) and the National Key Research and Development Program (2019YFC1907705).

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

This work was supported by the National Natural Science Foundation of China (22421003, 22378393, 22508388) and the National Key Research and Development Program (2019YFC1907705).

References

[1] Y. Xu, S. Cai, B. Chi, Z. Tu, Numerical study on improved mass and heat transfer performance in a solid oxide electrolysis cell with sine wave flow field, Int. J. Hydrogen Energy 137 (2025) 806–818, https://doi.org/10.1016/j.ijhydene.2024.04.266.

[2] J.E.T. Bistline, G.J. Blanford, Impact of carbon dioxide removal technologies on deep decarbonization of the electric power sector, Nat. Commun. 12 (2021) 3732, https://doi.org/10.1038/s41467-021-23554-6.

[3] Y. Xu, S. Cai, B. Chi, Z. Tu, Optimal design and performance enhancement based on field synergy theory in a solid oxide electrolysis cell with blockage flow channel, Electrochim. Acta 479 (2024) 143842, https://doi.org/10.1016/j.electacta.2024.143842.

[4] M.K.G. Deshmukh, M. Sameeroddin, D. Abdul, M. Abdul Sattar, Renewable energy in the 21st century: a review, Mater. Today Proc. 80 (2023) 1756–1759, https://doi.org/10.1016/j.matpr.2021.05.501.

[5] Y. Xu, S. Cai, B. Chi, Z. Tu, Technological limitations and recent developments in a solid oxide electrolyzer cell: a review, Int. J. Hydrogen Energy 50 (2024) 548–591, https://doi.org/10.1016/j.ijhydene.2023.08.314.

[6] R. Detz, M. Beerse, N. Meulendijks, P. Buskens, B. Van Der Zwaan, Towards the use of renewable syngas for the decarbonization of industry, ChemSusChem 17 (2024) e202400059, https://doi.org/10.1002/cssc.202400059.

[7] L. Zhang, F. Ma, Q. Zeng, F. Peng, Q. Hu, K. Wu, Y. Yang, H. Lu, B. Liang, Direct methanation of CO_2 in biogas with hydrogen from water electrolysis: the catalyst and system efficiency, Energy Fuels 36 (2022) 4416–4426, https://doi.org/10.1021/acs.energyfuels.2c00437.

[8] S. Biswas, A.P. Kulkarni, S. Giddey, S. Bhattacharya, A review on synthesis of methane as a pathway for renewable energy storage with a focus on solid oxide electrolytic cell-based processes, Front. Energy Res. 8 (2020) 570112, https://doi.org/10.3389/fenrg.2020.570112.

[9] X. Hou, Y. Jiang, K. Wei, C. Jiang, T.-C. Jen, Y. Yao, X. Liu, J. Ma, J.T.S. Irvine, Syngas production from CO_2 and H_2O via solid-oxide electrolyzer cells: fundamentals, materials, degradation, operating conditions, and applications, Chem. Rev. 124 (2024) 5119–5166, https://doi.org/10.1021/acs.chemrev.3c00760.

[10] K. Hong, S.N. Sutanto, J.A. Lee, J. Hong, Ni-based bimetallic nano-catalysts anchored on $BaZr_{0.4}Ce_{0.4}Y_{0.1}Yb_{0.1}O_{3-\delta}$ for internal steam reforming of methane in a low-temperature proton-conducting ceramic fuel cell, J. Mater. Chem. A 9 (2021) 6139–6151, https://doi.org/10.1039/D0TA11359J.

[11] L. Lei, J. Zhang, Z. Yuan, J. Liu, M. Ni, F. Chen, Progress report on proton conducting solid oxide electrolysis cells, Adv. Funct. Mater. 29 (2019) 1903805, https://doi.org/10.1002/adfm.201903805.

[12] Z. Pan, J. Wang, L. Zhu, C. Duan, Z. Jiao, Z. Zhong, R. O'Hayre, N.P. Sullivan, Performance and stability of renewable fuel production via H_2O electrolysis and H_2O–CO_2 co-electrolysis using proton-conducting solid oxide electrolysis cells, Appl. Energy 385 (2025) 125571, https://doi.org/10.1016/j.apenergy.2025.125571.

[13] Y. Ye, W. Lee, J. Pan, X. Sun, M. Zhou, J. Li, N. Zhang, J.W. Han, Y. Chen, Tuning the product selectivity of CO_2/H_2O co-electrolysis using CeO_2 -modified proton-conducting electrolysis cells, Energy Environ. Sci. 16 (2023) 3137–3145, https://doi.org/10.1039/D3EE01468A.

[14] N. Shi, Y. Xie, D. Huan, Y. Yang, S. Xue, Z. Qi, Y. Pan, R. Peng, C. Xia, Y. Lu, Controllable CO_2 conversion in high performance proton conducting solid oxide electrolysis cells and the possible mechanisms, J. Mater. Chem. A 7 (2019) 4855–4864, https://doi.org/10.1039/C8TA12458B.

[15] P. Błaszczak, M. Zając, A. Ducka, K. Matlak, B. Wolanin, S.-F. Wang, A. Mandziak, B. Bochentyn, P. Jasiński, High-temperature Co-electrolysis of CO_2/H_2O and direct methanation over Co-impregnated SOEC. Bimetallic synergy between Co and Ni, Int. J. Hydrogen Energy 47 (2022) 35017–35037, https://doi.org/10.1016/j.ijhydene.2022.08.057.

[16] Y. Chen, Y. Luo, Y. Shi, N. Cai, Theoretical modeling of a pressurized tubular reversible solid oxide cell for methane production by co-electrolysis, Appl. Energy 268 (2020) 114927, https://doi.org/10.1016/j.apenergy.2020.114927.

[17] Y. Luo, Y. Shi, W. Li, N. Cai, Synchronous enhancement of H_2O/CO_2 co-electrolysis and methanation for efficient one-step power-to-methane, Energy Convers. Manag. 165 (2018) 127–136, https://doi.org/10.1016/j.enconman.2018.03.028.

[18] Y. Luo, W. Li, Y. Shi, T. Cao, X. Ye, S. Wang, N. Cai, Experimental characterization and theoretical modeling of methane production by H_2O/CO_2 Co-Electrolysis in a

tubular solid oxide electrolysis cell, J. Electrochem. Soc. 162 (2015) F1129–F1134, https://doi.org/10.1149/2.0171510jes.
[19] S.H. Jensen, J.V.T. Høgh, R. Barfod, M.B. Mogensen, High temperature electrolysis of steam and carbon dioxide, Risø Int. Energy Conf. 2003 (2003) 204–215.
[20] C. Stoots, J. O'Brien, J. Hartvigsen, Results of recent high temperature coelectrolysis studies at the Idaho national laboratory, Int. J. Hydrogen Energy 34 (2009) 4208–4215, https://doi.org/10.1016/j.ijhydene.2008.08.029.
[21] S. Tian, X. Yang, X. Chen, G. Li, A. Aikelaimu, Y. Meng, Y. Gao, C. Lang, M. Fan, Catalytic calcium-looping reforming of biogas: a novel strategy to produce syngas with improved H_2/CO molar ratios, J. Clean. Prod. 270 (2020) 122504, https://doi.org/10.1016/j.jclepro.2020.122504.
[22] Y. Wang, T. Liu, S. Fang, G. Xiao, H. Wang, F. Chen, A novel clean and effective syngas production system based on partial oxidation of methane assisted solid oxide co-electrolysis process, J. Power Sources 277 (2015) 261–267, https://doi.org/10.1016/j.jpowsour.2014.11.092.
[23] R. Elder, D. Cumming, M.B. Mogensen, High temperature electrolysis, in: Carbon Dioxide Util, Elsevier, 2015, pp. 183–209, https://doi.org/10.1016/B978-0-444-62746-9.00011-6.
[24] W. Li, Y. Shi, Y. Luo, N. Cai, Elementary reaction modeling of CO_2/H_2O co-electrolysis cell considering effects of cathode thickness, J. Power Sources 243 (2013) 118–130, https://doi.org/10.1016/j.jpowsour.2013.05.119.
[25] J. Gao, Y. Wang, Y. Ping, D. Hu, G. Xu, F. Gu, F. Su, A thermodynamic analysis of methanation reactions of carbon oxides for the production of synthetic natural gas, RSC Adv. 2 (2012) 2358, https://doi.org/10.1039/c2ra00632d.
[26] K. Xie, Y. Zhang, G. Meng, J.T.S. Irvine, Direct synthesis of methane from CO_2/H_2O in an oxygen-ion conducting solid oxide electrolyser, Energy Environ. Sci. 4 (2011) 2218, https://doi.org/10.1039/c1ee01035b.
[27] W. Li, H. Wang, Y. Shi, N. Cai, Performance and methane production characteristics of H_2O–CO_2 co-electrolysis in solid oxide electrolysis cells, Int. J. Hydrogen Energy 38 (2013) 11104–11109, https://doi.org/10.1016/j.ijhydene.2013.01.008.
[28] D.M. Bierschenk, J.R. Wilson, S.A. Barnett, High efficiency electrical energy storage using a methane–oxygen solid oxide cell, energy env, Science 4 (2011) 944–951, https://doi.org/10.1039/C0EE00457J.
[29] L. Lei, T. Liu, S. Fang, J.P. Lemmon, F. Chen, The co-electrolysis of CO_2–H_2O to methane via a novel micro-tubular electrochemical reactor, J. Mater. Chem. A 5 (2017) 2904–2910, https://doi.org/10.1039/C6TA10252B.
[30] L. Chen, F. Chen, C. Xia, Direct synthesis of methane from CO_2–H_2O co-electrolysis in tubular solid oxide electrolysis cells, energy env, Science 7 (2014) 4018–4022, https://doi.org/10.1039/C4EE02786H.
[31] S. Zong, X. Zhao, L.L. Jewell, Y. Zhang, X. Liu, Advances and challenges with SOEC high temperature co-electrolysis of CO_2/H_2O: materials development and technological design, Carbon Capture Sci. Technol. 12 (2024) 100234, https://doi.org/10.1016/j.ccst.2024.100234.
[32] B. Chen, H. Xu, M. Ni, Modelling of SOEC-FT reactor: pressure effects on methanation process, Appl. Energy 185 (2017) 814–824, https://doi.org/10.1016/j.apenergy.2016.10.095.
[33] J. Kuang, Y. Shi, N. Cai, H. Wang, W. Li, Reaction characteristics of hydrocarbon production by H_2O-CO_2 Co-electrolysis in solid oxide electrolysis cells, in Chinese, Proc. CSEE 32 (2012) 31–35, https://doi.org/10.13334/j.0258-8013.pcsee.2012.17.005.
[34] Y. Luo, Y. Shi, Y. Chen, W. Li, L. Jiang, N. Cai, Pressurized tubular solid oxide H_2O/CO_2 coelectrolysis cell for direct power-to-methane, AIChE J. 66 (2020) e16896, https://doi.org/10.1002/aic.16896.
[35] X. Zhou, X. Pan, H. Yang, T. Li, Z. Lyu, D. Cui, S. Wu, Status and outlook of solid oxide cells for hydrocarbon fuel conversion, ChemSusChem 18 (2025) e202501131, https://doi.org/10.1002/cssc.202501131.
[36] Osaka Gas Co., Ltd., Osaka Gas Sel. Green Innov. Fund Proj, SOEC methanation-challenging dev. Worlds Most effic. Synth. Methane prod. Technol. https://www.osakagas.co.jp/company/press/pr2022/1305922_49634.html, 2022. (Accessed 25 November 2025).
[37] Y. Naumovich, S. Jagielski, J. Kupecki, K. Motylinski, K. Martsinchyk, A. Niemczyk, J. Milewski, Carbon dioxide capture by direct methanation in co-electrolysis using solid oxide cell, Sustain. Mater. Technol. 40 (2024) e00944, https://doi.org/10.1016/j.susmat.2024.e00944.
[38] E. Le Saché, L. Pastor-Pérez, B.J. Haycock, J.J. Villora-Pico, A. Sepúlveda-Escribano, T.R. Reina, Switchable catalysts for chemical CO_2 recycling: a step forward in the methanation and reverse water–gas shift reactions, ACS Sustain. Chem. Eng. 8 (2020) 4614–4622.
[39] M. Liu, M.E. Lynch, K. Blinn, F.M. Alamgir, Y. Choi, Rational SOFC material design: new advances and tools, mater, Today Off. 14 (2011) 534–546, https://doi.org/10.1016/S1369-7021(11)70279-6.
[40] T. Hu, F. He, M. Liu, Y. Chen, In situ/operando regulation of the reaction activities on hetero-structured electrodes for solid oxide cells, Prog. Mater. Sci. 133 (2023) 101050, https://doi.org/10.1016/j.pmatsci.2022.101050.
[41] T.L. Skafte, Z. Guan, M.L. Machala, C.B. Gopal, M. Monti, L. Martinez, E. Stamate, S. Sanna, J.A. Garrido Torres, E.J. Crumlin, M. García-Melchor, M. Bajdich, W. C. Chueh, C. Graves, Selective high-temperature CO_2 electrolysis enabled by oxidized carbon intermediates, Nat. Energy 4 (2019) 846–855, https://doi.org/10.1038/s41560-019-0457-4.
[42] Y. Yang, Y. Li, Y. Jiang, M. Zheng, T. Hong, X. Wu, C. Xia, The electrochemical performance and CO_2 reduction mechanism on strontium doped lanthanum ferrite fuel electrode in solid oxide electrolysis cell, Electrochim. Acta 284 (2018) 159–167, https://doi.org/10.1016/j.electacta.2018.07.187.
[43] Z.A. Feng, M.L. Machala, W.C. Chueh, Surface electrochemistry of CO_2 reduction and CO oxidation on Sm-doped CeO_{2-x}: coupling between Ce^{3+} and carbonate adsorbates, Phys. Chem. Chem. Phys. 17 (2015) 12273–12281, https://doi.org/10.1039/C5CP00114E.
[44] W. Li, Y. Shi, Y. Luo, Y. Wang, N. Cai, Carbon deposition on patterned nickel/yttria stabilized zirconia electrodes for solid oxide fuel cell/solid oxide electrolysis cell modes, J. Power Sources 276 (2015) 26–31, https://doi.org/10.1016/j.jpowsour.2014.11.106.
[45] P. Kim-Lohsoontorn, J. Bae, Electrochemical performance of solid oxide electrolysis cell electrodes under high-temperature coelectrolysis of steam and carbon dioxide, J. Power Sources 196 (2011) 7161–7168, https://doi.org/10.1016/j.jpowsour.2010.09.018.
[46] J. Hartvigsen, S. Elangovan, L. Frost, A. Nickens, C.M. Stoots, J.E. O'Brien, J. S. Herring, Carbon dioxide recycling by high temperature Co-electrolysis and hydrocarbon synthesis, ECS Trans. 12 (2008) 625–637, https://doi.org/10.1149/1.2921588.
[47] M. Riegraf, A. Surrey, R. Costa, Ni/CGO-Based electrolyte-supported cells during co-electrolysis: the role of feed gas impurities, ECS Trans. 103 (2021) 1129–1135, https://doi.org/10.1149/10301.1129ecst.
[48] S.D. Ebbesen, C. Graves, M. Mogensen, Production of synthetic fuels by Co-Electrolysis of steam and carbon dioxide, Int. J. Green Energy 6 (2009) 646–660, https://doi.org/10.1080/15435070903372577.
[49] C.M. Stoots, J.E. O'Brien, J.S. Herring, J.J. Hartvigsen, Syngas production via high-temperature coelectrolysis of steam and carbon dioxide, J. Fuel Cell Sci. Technol. 6 (2009) 011014, https://doi.org/10.1115/1.2971061.
[50] G. Rabenstein, V. Hacker, Hydrogen for fuel cells from ethanol by steam-reforming, partial-oxidation and combined auto-thermal reforming: a thermodynamic analysis, J. Power Sources 185 (2008) 1293–1304, https://doi.org/10.1016/j.jpowsour.2008.08.010.
[51] C. Cui, Y. Wang, Y. Tong, S. Wang, C. Chen, Z. Zhan, Syngas production through CH_4-assisted co-electrolysis of H_2O and CO_2 in $La_{0.8}Sr_{0.2}Cr_{0.5}Fe_{0.5}O_{3-\delta}$-$Zr_{0.84}Y_{0.16}O_{2-\delta}$ electrode-supported solid oxide electrolysis cells, Int. J. Hydrogen Energy 46 (2021) 20305–20312, https://doi.org/10.1016/j.ijhydene.2021.03.177.
[52] N. Fujiwara, R. Kikuchi, A. Takagaki, S.T. Oyama, Development OF solid oxide electrolysis cell cathodes for direct methanation IN CO_2/H_2O CO-electrolysis, Gd, Renew. Energy Proc. Jpn. Counc. Renew. Energy 2018 (2018) 225.
[53] H. Zheng, Y. Tian, L. Zhang, B. Chi, J. Pu, L. Jian, $La_{0.8}Sr_{0.2}Co_{0.8}Ni_{0.2}O_{3-\delta}$ impregnated oxygen electrode for H_2O/CO_2 co-electrolysis in solid oxide electrolysis cells, J. Power Sources 383 (2018) 93–101, https://doi.org/10.1016/j.jpowsour.2018.02.041.
[54] S.E. Wolf, L. Dittrich, M. Nohl, T. Duyster, I.C. Vinke, R.-A. Eichel, L.G.J. (Bert) De Haart, Boundary investigation of high-temperature Co-Electrolysis towards direct CO_2 electrolysis, J. Electrochem. Soc. 169 (2022) 034531, https://doi.org/10.1149/1945-7111/ac5e45.
[55] M. Preininger, B. Stoeckl, V. Subotić, F. Mittmann, C. Hochenauer, Performance of a ten-layer reversible Solid Oxide cell stack (rSOC) under transient operation for autonomous application, Appl. Energy 254 (2019) 113695, https://doi.org/10.1016/j.apenergy.2019.113695.
[56] T. Yamaguchi, H. Shimada, U. Honda, H. Kishimoto, T. Ishiyama, Y. Fujishiro, Development of electrochemical methanation reactor with Co-Electrolysis of humidified CO_2, ECS Trans. 68 (2015) 3459–3463, https://doi.org/10.1149/06801.3459ecst.
[57] K. Wakamatsu, T. Yasuda, M. Aratani, T. Ogura, Theoretical study of catalytic activity modifications in CO_2 methanation induced by an electric field in solid-oxide cells, Catal. Sci. Technol. 14 (2024) 704–717, https://doi.org/10.1039/D3CY01327H.
[58] Q. Li, Y. Zheng, Y. Sun, T. Li, C. Xu, W. Wang, S.H. Chan, Understanding the occurrence of the individual CO_2 electrolysis during H_2O-CO_2 co-electrolysis in classic planar Ni-YSZ/YSZ/LSM-YSZ solid oxide cells, Electrochim. Acta 318 (2019) 440–448, https://doi.org/10.1016/j.electacta.2019.06.108.
[59] J. Aicart, J. Laurencin, M. Petitjean, L. Dessemond, Experimental validation of two-dimensional H_2O and CO_2 co-electrolysis modeling, Fuel Cells 14 (2014) 430–447, https://doi.org/10.1002/fuce.201300214.
[60] C. Graves, S.D. Ebbesen, M. Mogensen, Co-electrolysis of CO_2 and H_2O in solid oxide cells: performance and durability, Solid State Ionics 192 (2011) 398–403, https://doi.org/10.1016/j.ssi.2010.06.014.
[61] Y. Wang, C. Wu, B. Zu, M. Han, Q. Du, M. Ni, K. Jiao, Ni migration of Ni-YSZ electrode in solid oxide electrolysis cell: an integrated model study, J. Power Sources 516 (2021) 230660, https://doi.org/10.1016/j.jpowsour.2021.230660.
[62] S.D. Ebbesen, J.B. Hansen, M. Mogensen, Biogas upgrading using SOEC with a Ni-ScYSZ electrode, ECS Trans. 57 (2013) 3217–3227, https://doi.org/10.1149/05701.3217ecst.
[63] G. Horiguchi, T. Yamaguchi, H. Tateno, K.D. Bagarinao, H. Kishimoto, T. Mochizuki, Preparation of Ni/YSZ catalysts for application of solid oxide electrolysis cell methanation, ECS Trans. 111 (2023) 357–361, https://doi.org/10.1149/11106.0357ecst.
[64] J.F.M. Simons, T.J. De Heer, R.C.J. Van De Poll, V. Muravev, N. Kosinov, E.J. M. Hensen, Structure sensitivity of CO_2 hydrogenation on Ni revisited, J. Am. Chem. Soc. 145 (2023) 20289–20301, https://doi.org/10.1021/jacs.3c04284.
[65] Y. Du, L. Zhao, Y. Xiao, J. Kong, P. Liu, X. Yang, T. Zhou, High electrolysis performance of the SOEC cathode by creating oxygen vacancies to regulate the adsorption energy, J. Mater. Chem. A 12 (2024) 28911–28918, https://doi.org/10.1039/D4TA04915B.

[66] Y. Jiang, F. Chen, C. Xia, A review on cathode processes and materials for electroreduction of carbon dioxide in solid oxide electrolysis cells, J. Power Sources 493 (2021) 229713, https://doi.org/10.1016/j.jpowsour.2021.229713.

[67] N. Li, M. Wang, Q. Shen, Y. Teng, D. Wang, C. Chen, Z. Zhan, Reduced concentration polarization and enhanced steam throughput conversion with a solid oxide electrolysis cell supported on an electrode with optimized pore structure, Int. J. Hydrogen Energy 47 (2022) 21673–21680, https://doi.org/10.1016/j.ijhydene.2022.05.013.

[68] D. Dong, S. Xu, X. Shao, L. Hucker, J. Marin, T. Pham, K. Xie, Z. Ye, P. Yang, L. Yu, G. Parkinson, C.-Z. Li, Hierarchically ordered porous Ni-based cathode-supported solid oxide electrolysis cells for stable CO_2 electrolysis without safe gas, J. Mater. Chem. A 5 (2017) 24098–24102, https://doi.org/10.1039/C7TA06839E.

[69] S. Gu, S. Hu, C. Yang, Z. Wang, J. Pu, B. Chi, Ni-SDC nanoparticles decorated Ni-YSZ fuel electrode for SOEC: a comprehensive strategy to optimize performance, syngas composition and stability for H_2O-CO_2 co-electrolysis, Fuel 389 (2025) 134600, https://doi.org/10.1016/j.fuel.2025.134600.

[70] S.-W. Kim, M. Park, H. Kim, K.J. Yoon, J.-W. Son, J.-H. Lee, B.-K. Kim, J.-H. Lee, J. Hong, In-situ nano-alloying Pd-Ni for economical control of syngas production from high-temperature thermo-electrochemical reduction of steam/CO_2, Appl. Catal. B Environ. 200 (2017) 265–273, https://doi.org/10.1016/j.apcatb.2016.07.008.

[71] H. Jeong, S. Kim, Y. Bae, K.J. Yoon, J. Lee, J. Hong, Effect of Fe infiltration to Ni/YSZ solid-oxide-cell fuel electrode on steam/CO_2 co-electrolysissis, Int. J. Energy Res. 43 (2019) 4949–4958, https://doi.org/10.1002/er.4577.

[72] M. Zheng, S. Wang, Y. Yang, C. Xia, Barium carbonate as a synergistic catalyst for the H_2O/CO_2 reduction reaction at Ni–yttria stabilized zirconia cathodes for solid oxide electrolysis cells, J. Mater. Chem. A 6 (2018) 2721–2729, https://doi.org/10.1039/C7TA08249E.

[73] A.P. Kulkarni, S. Giddey, S.P.S. Badwal, Efficient conversion of CO_2 in solid oxide electrolytic cells with Pd doped perovskite cathode on ceria nanofilm interlayer, J. CO_2 Util. 17 (2017) 180–187, https://doi.org/10.1016/j.jcou.2016.11.014.

[74] X. Yue, J.T.S. Irvine, Modification of LSCM–GDC cathodes to enhance performance for high temperature CO_2 electrolysis using solid oxide electrolysis cells (SOECs), J. Mater. Chem. A 5 (2017) 7081–7090, https://doi.org/10.1039/C6TA09421J.

[75] X. Yang, K. Sun, M. Ma, C. Xu, R. Ren, J. Qiao, Z. Wang, S. Zhen, R. Hou, W. Sun, Achieving strong chemical adsorption ability for efficient carbon dioxide electrolysis, Appl. Catal. B Environ. 272 (2020) 118968, https://doi.org/10.1016/j.apcatb.2020.118968.

[76] S. Wang, B. Qian, Z. Wang, B. Yin, Y. Zheng, L. Ge, H. Chen, H. Yang, High catalytic activity of Fe-based perovskite fuel electrode for direct CO_2 electroreduction in SOECs, J. Alloys Compd. 888 (2021) 161573, https://doi.org/10.1016/j.jallcom.2021.161573.

[77] D.J. Cumming, A.R. Thompson, R.H. Rothman, Nickel impregnated cerium-doped strontium titanate fuel electrode: direct carbon dioxide electrolysis and Co-Electrolysis, J. Electrochem. Soc. 163 (2016) F3057–F3061, https://doi.org/10.1149/2.0081611jes.

[78] C. Jin, C. Yang, F. Zhao, D. Cui, F. Chen, $La_{0.75}Sr_{0.25}Cr_{0.5}Mn_{0.5}O_3$ as hydrogen electrode for solid oxide electrolysis cells, Int. J. Hydrogen Energy 36 (2011) 3340–3346, https://doi.org/10.1016/j.ijhydene.2010.12.085.

[79] Y. Alizad Farzin, M. Bjerg Mogensen, S. Pirou, H. Lund Frandsen, Perovskite/ruddlesden-popper composite fuel electrode of strontium-praseodymium-manganese oxide for solid oxide cells: an alternative candidate, J. Power Sources 580 (2023) 233450, https://doi.org/10.1016/j.jpowsour.2023.233450.

[80] H. Lv, Y. Zhou, X. Zhang, Y. Song, Q. Liu, G. Wang, X. Bao, Infiltration of $Ce_{0.8}Gd_{0.2}O_{1.9}$ nanoparticles on $Sr_2Fe_{1.5}Mo_{0.5}O_6$- cathode for CO_2 electroreduction in solid oxide electrolysis cell, J. Energy Chem. 35 (2019) 71–78, https://doi.org/10.1016/j.jechem.2018.11.002.

[81] Q. Liu, C. Yang, X. Dong, F. Chen, Perovskite $Sr_2Fe_{1.5}Mo_{0.5}O_{6-\delta}$ as electrode materials for symmetrical solid oxide electrolysis cells, Int. J. Hydrogen Energy 35 (2010) 10039–10044, https://doi.org/10.1016/j.ijhydene.2010.08.016.

[82] C. Li, Y. Deng, L. Yang, B. Liu, D. Yan, L. Fan, J. Li, L. Jia, An active and stable hydrogen electrode of solid oxide cells with exsolved fe–co–ni nanoparticles from $Sr_2FeCo_{0.2}Ni_{0.2}Mo_{0.6}O_{6-\delta}$ double-perovskite, Adv. Powder Mater. 2 (2023) 100133, https://doi.org/10.1016/j.apmate.2023.100133.

[83] L. Gan, L. Ye, S. Wang, M. Liu, S. Tao, K. Xie, Demonstration of direct conversion of CO_2/H_2O into syngas in a symmetrical proton-conducting solid oxide electrolyzer, Int. J. Hydrogen Energy 41 (2016) 1170–1175, https://doi.org/10.1016/j.ijhydene.2015.11.032.

[84] M. Yang, Z. Yao, S. Liu, J. Wang, A. Sun, H. Xu, G. Yang, R. Ran, W. Zhou, G. Xiao, Z. Shao, Bismuth doped $Sr_2Fe_{1.5}Mo_{0.5}O_6$- double perovskite as a robust fuel electrode in ceramic oxide cells for direct CO_2 electrolysis, J. Mater. Sci. Technol. 164 (2023) 160–167, https://doi.org/10.1016/j.jmst.2023.04.061.

[85] S. Liu, M. Yang, R. Xu, X. Xiang, G. Yang, H. Xu, G. Xiao, R. Ran, W. Zhou, Z. Shao, *In situ* passivation of Fe nanoparticles exsolved from perovskite cathodes through zinc doping for CO_2 electrolysis, Green Chem. 25 (2023) 9826–9836, https://doi.org/10.1039/D3GC03518B.

[86] S. Liu, Q. Liu, J.-L. Luo, Highly stable and efficient catalyst with in situ exsolved Fe–Ni alloy nanospheres socketed on an oxygen deficient perovskite for direct CO_2 electrolysis, ACS Catal. 6 (2016) 6219–6228, https://doi.org/10.1021/acscatal.6b01555.

[87] D. Liu, H. Shang, C. Zhou, J. Miao, D. Xue, Z. Chen, M. Fei, F. Liang, Q. Niu, R. Ran, W. Zhou, Z. Shao, Active Cu and Fe nanoparticles codecorated Ruddlesden–Popper-Type perovskite as solid oxide electrolysis cells cathode for CO_2 splitting, Energy Environ. Mater. 7 (2024) e12717, https://doi.org/10.1002/eem2.12717.

[88] Y. Zhu, N. Zhang, W. Zhang, Y. Gong, R. Wang, H. Wang, J. Jin, L. Zhao, B. He, Probing metal/high-entropy perovskite heterointerfaces for efficient and sustainable CO_2 electroreduction, J. Mater. Chem. A 12 (2024) 18182–18192, https://doi.org/10.1039/D4TA02372B.

[89] Z. Wang, T. Tan, K. Du, Q. Zhang, M. Liu, C. Yang, A high-entropy layered perovskite coated with in situ exsolved core-shell CuFe@FeOx nanoparticles for efficient CO_2 electrolysis, Adv. Mater. 36 (2024) 2312119, https://doi.org/10.1002/adma.202312119.

[90] J. Qi, L. Bian, T. Ting, C. Liu, L. Yang, Y. Xu, J. Peng, X. Song, S. An, Boosting electrochemical CO_2 directly electrolysis by tuning the surface oxygen defect of perovskite, J. Power Sources 570 (2023) 233032, https://doi.org/10.1016/j.jpowsour.2023.233032.

[91] A. Blanco, J. Caroca, R. Tamayo, M. Flores, M. Romero-Sáez, R. Espinoza-González, F. Gracia, CO_2 methanation activity of Ni-doped perovskites, Fuel 320 (2022) 123954, https://doi.org/10.1016/j.fuel.2022.123954.

[92] P. Steiger, O. Kröcher, D. Ferri, Increased nickel exsolution from $LaFe_{0.8}Ni_{0.2}O_3$ perovskite-derived CO_2 methanation catalysts through strontium doping, Appl. Catal. Gen. 590 (2020) 117328, https://doi.org/10.1016/j.apcata.2019.117328.

[93] J.C. Pedrozo Alfonso, F.G. Duran, L.E. Cadús, O.J. Furlong, M.S. Nazzarro, A. M. Beltrán, G.M. Arzac, A. Fernández, F.N. Aguero, In situ generation of Ni particles from $CaTi_{1-x}Ni_xO_3$ perovskites used in CO_2 methanation, Appl. Catal. Gen. 709 (2026) 120622, https://doi.org/10.1016/j.apcata.2025.120622.

[94] H. Arandiyan, Y. Wang, J. Scott, S. Mesgari, H. Dai, R. Amal, In situ exsolution of Bimetallic Rh–Ni nanoalloys: a highly efficient catalyst for CO_2 methanation, ACS Appl. Mater. Interfaces 10 (2018) 16352–16357, https://doi.org/10.1021/acsami.8b00889.

[95] A.A. Alkhoori, E.P. Komarala, A.A. Dabbawala, A.G.S. Hussien, D.H. Anjum, S. Mao, K. Polychronopoulou, Exsolved $LaNiRuO_3$ perovskite-based catalysts for CO_2 methanation reaction, RSC Sustain. 2 (2024) 3866–3878, https://doi.org/10.1039/D4SU00410H.

[96] S. Zhang, Y. Jiang, H. Han, Y. Li, C. Xia, Perovskite oxyfluoride ceramic with in situ exsolved Ni–Fe nanoparticles for direct CO_2 electrolysis in solid oxide electrolysis cells, ACS Appl. Mater. Interfaces 14 (2022) 28854–28864, https://doi.org/10.1021/acsami.2c05324.

[97] Y. Guo, S. Wang, R. Li, J. Yu, X. Zhang, M. Li, X. Zheng, J. Zhu, Y. Song, G. Wang, X. Bao, In situ exsolved CoFe alloy nanoparticles for stable anodic methane reforming in solid oxide electrolysis cells, Joule 8 (2024) 2016–2032, https://doi.org/10.1016/j.joule.2024.04.009.

[98] X. Wang, Y. Zhao, J. Feng, P. Wang, X. Feng, K. Yang, X. Ding, In situ exsolved bimetallic nanoparticles on double perovskite for efficient CO_2 electrolysis and synergistic methane reforming in SOEC, Small 21 (2025) 2505492, https://doi.org/10.1002/smll.202505492.

[99] M. Noponen, P. Torri, J. Göös, J. Puranen, T. Lehtinen, S. Pylypko, E. Ounpuu, Progress of SOC development at elcogen, ECS Trans. 103 (2021) 267–274, https://doi.org/10.1149/10301.0267ecst.

[100] Solid oxide electrolysis cells, afford. Green hydrog. https://elcogen.com/products/solid-oxide-electrolysis-cells/. (Accessed 23 November 2025).

[101] M. Lo Faro, S.C. Zignani, S. Trocino, V. Antonucci, A.S. Aricò, New insights on the co-electrolysis of CO_2 and H_2O through a solid oxide electrolyser operating at intermediate temperatures, Electrochim. Acta 296 (2019) 458–464, https://doi.org/10.1016/j.electacta.2018.11.079.

[102] M. Lo Faro, W. Oliveira Da Silva, W. Valenzuela Barrientos, G.G.A. Saglietti, S. C. Zignani, E.A. Ticianelli, V. Antonucci, A.S. Aricò, The role of CuSn alloy in the co-electrolysis of CO_2 and H_2O through an intermediate temperature solid oxide electrolyser, J. Energy Storage 27 (2020) 100820, https://doi.org/10.1016/j.est.2019.100820.

[103] M. Lo Faro, S. Campagna Zignani, V. Antonucci, A.S. Aricò, The effect of Ni-Modified LSFCO promoting layer on the gas produced through Co-electrolysis of CO_2 and H_2O at intermediate temperatures, Catalysts 11 (2021) 56, https://doi.org/10.3390/catal11010056.

[104] M. Lo Faro, W. Oliveira Da Silva, W.V. Barrientos, G.G.A. Saglietti, S.C. Zignani, V. Antonucci, E.A. Ticianelli, A.S. Aricò, Enhanced production of methane through the use of a catalytic Ni–Fe pre-layer in a solid oxide co-electrolyser, Int. J. Hydrogen Energy 45 (2020) 5134–5142, https://doi.org/10.1016/j.ijhydene.2019.06.161.

[105] E. Ruiz, J. Aldecoa, Á. Morales, M. Farchado, J.M. Sánchez, Methanation of CO_2 on Cu in a tubular co-ionic SOEC, Int. J. Hydrogen Energy 52 (2024) 1338–1359, https://doi.org/10.1016/j.ijhydene.2023.08.325.

[106] H. Liu, S. Xu, G. Zhou, K. Xiong, Z. Jiao, S. Wang, CO_2 hydrogenation to methane over Co/KIT-6 catalysts: effect of Co content, Fuel 217 (2018) 570–576, https://doi.org/10.1016/j.fuel.2017.12.112.

[107] I. Czekaj, F. Loviat, F. Raimondi, J. Wambach, S. Biollaz, A. Wokaun, Characterization of surface processes at the Ni-based catalyst during the methanation of biomass-derived synthesis gas: X-ray photoelectron spectroscopy (XPS), Appl. Catal. Gen. 329 (2007) 68–78, https://doi.org/10.1016/j.apcata.2007.06.027.

[108] M. Tao, X. Meng, Y. Lv, Z. Bian, Z. Xin, Effect of impregnation solvent on Ni dispersion and catalytic properties of Ni/SBA-15 for CO methanation reaction, Fuel 165 (2016) 289–297, https://doi.org/10.1016/j.fuel.2015.10.023.

[109] S. De, J. Zhang, R. Luque, N. Yan, Ni-based bimetallic heterogeneous catalysts for energy and environmental applications, Energy Environ. Sci. 9 (2016) 3314–3347, https://doi.org/10.1039/C6EE02002J.

[110] P. Błaszczak, A. Mizera, B. Bochentyn, S.-F. Wang, P. Jasiński, Preparation of methanation catalysts for high temperature SOEC by β-cyclodextrin-assisted impregnation of nano-CeO_2 with transition metal oxides, Int. J. Hydrogen Energy 47 (2022) 1901–1916, https://doi.org/10.1016/j.ijhydene.2021.10.144.

[111] A.H. Hatta, A.A. Jalil, N.S. Hassan, M.Y.S. Hamid, A.F.A. Rahman, L.P. Teh, D. Prasetyoko, A review on recent bimetallic catalyst development for synthetic natural gas production via CO methanation, Int. J. Hydrogen Energy 47 (2022) 30981–31002, https://doi.org/10.1016/j.ijhydene.2021.10.213.

[112] C. Mebrahtu, F. Krebs, S. Perathoner, S. Abate, G. Centi, R. Palkovits, Hydrotalcite based Ni–Fe/(Mg, Al)O_x catalysts for CO_2 methanation – tailoring Fe content for improved CO dissociation, basicity, and particle size, Catal. Sci. Technol. 8 (2018) 1016–1027, https://doi.org/10.1039/C7CY02099F.

[113] D. Pandey, G. Deo, Promotional effects in alumina and silica supported bimetallic Ni–Fe catalysts during CO_2 hydrogenation, J. Mol. Catal. Chem. 382 (2014) 23–30, https://doi.org/10.1016/j.molcata.2013.10.022.

[114] A.I. Tsiotsias, N.D. Charisiou, I.V. Yentekakis, M.A. Goula, Bimetallic Ni-Based catalysts for CO_2 methanation: a review, Nanomaterials 11 (2020) 28, https://doi.org/10.3390/nano11010028.

[115] S. Biswas, A.P. Kulkarni, D. Fini, S. Singh Rathore, A. Seeber, S. Giddey, S. Bhattacharya, Catalyst-induced enhancement of direct methane synthesis in solid oxide electrolyser, Electrochim. Acta 391 (2021) 138934, https://doi.org/10.1016/j.electacta.2021.138934.

[116] S. Biswas, A.P. Kulkarni, D. Fini, S. Giddey, S. Bhattacharya, *In situ* synthesis of methane using Ag–GDC composite electrodes in a tubular solid oxide electrolytic cell: new insight into the role of oxide ion removal, Sustain. Energy Fuels 5 (2021) 2055–2064, https://doi.org/10.1039/D0SE01887B.

[117] N. Kokkinou, F. Xydas, S. Brosda, G. Kyriakou, A. Katsaounis, Electrochemical promotion of CO_2 hydrogenation using Rh catalysts supported on O^{2-} conducting solid electrolyte, Catalysts 13 (2023) 1014, https://doi.org/10.3390/catal13061014.

[118] A. Kotsiras, I. Kalaitzidou, D. Grigoriou, A. Symillidis, M. Makri, A. Katsaounis, C. G. Vayenas, Electrochemical promotion of nanodispersed Ru-Co catalysts for the hydrogenation of CO_2, Appl. Catal. B Environ. 232 (2018) 60–68, https://doi.org/10.1016/j.apcatb.2018.03.031.

[119] D. Theleritis, S. Souentie, A. Siokou, A. Katsaounis, C.G. Vayenas, Hydrogenation of CO_2 over Ru/YSZ electropromoted catalysts, ACS Catal. 2 (2012) 770–780, https://doi.org/10.1021/cs300072a.

[120] X. He, X. Huang, H. Sun, L. Gan, Enhanced CO_2 electrolysis with *in situ* exsolved nanoparticles in the perovskite cathode, New J. Chem. 48 (2024) 5834–5839, https://doi.org/10.1039/D4NJ00406J.

[121] J. Gan, N. Hou, T. Yao, L. Fan, T. Gan, Z. Huang, Y. Zhao, Y. Li, A high performing perovskite cathode with in situ exsolved Co nanoparticles for H_2O and CO_2 solid oxide electrolysis cell, Catal. Today 364 (2021) 89–96, https://doi.org/10.1016/j.cattod.2020.06.082.

[122] S.J. Baxter, M. Rine, B. Min, Y. Liu, J. Yao, Near 100% CO_2 conversion and CH_4 selectivity in a solid oxide electrolysis cell with integrated catalyst operating at 450°C, J. CO_2 Util. 59 (2022) 101954, https://doi.org/10.1016/j.jcou.2022.101954.

[123] J. Kim, Characterization of LSM–YSZ composite electrode by ac impedance spectroscopy, Solid State Ionics 143 (2001) 379–389, https://doi.org/10.1016/S0167-2738(01)00877-3.

[124] X. Chen, Electrochemical behavior of La(Sr)MnO_3 electrode under cathodic and anodic polarization, Solid State Ionics 167 (2004) 379–387, https://doi.org/10.1016/j.ssi.2003.08.049.

[125] K. Murakami, T. Matsui, R. Kikuchi, H. Muroyama, K. Eguchi, Activation of LSM electrode related to the potential oscillation under cathodic polarization, J. Electrochem. Soc. 157 (2010) B880, https://doi.org/10.1149/1.3374407.

[126] G. Kaur, A.P. Kulkarni, S. Giddey, S.P.S. Badwal, Ceramic composite cathodes for CO_2 conversion to CO in solid oxide electrolysis cells, Appl. Energy 221 (2018) 131–138, https://doi.org/10.1016/j.apenergy.2018.03.176.

[127] M.A. Laguna-Bercero, J.A. Kilner, S.J. Skinner, Development of oxygen electrodes for reversible solid oxide fuel cells with scandia stabilized zirconia electrolytes, Solid State Ionics 192 (2011) 501–504, https://doi.org/10.1016/j.ssi.2010.01.003.

[128] Y. Zheng, T. Chen, Q. Li, W. Wu, H. Miao, C. Xu, W.G. Wang, Achieving 360 NL h^{-1} hydrogen production rate through 30-Cell solid oxide electrolysis stack with LSCF–GDC composite oxygen electrode, Fuel Cells 14 (2014) 1066–1070, https://doi.org/10.1002/fuce.201400051.

[129] Y. Bo, Z. Wenqiang, X. Jingming, C. Jing, Microstructural characterization and electrochemical properties of $Ba_{0.5}Sr_{0.5}Co_{0.8}Fe_{0.2}O_{3-}$ and its application for anode of SOEC, Int. J. Hydrogen Energy 33 (2008) 6873–6877, https://doi.org/10.1016/j.ijhydene.2008.07.066.

[130] M.A. Laguna-Bercero, N. Kinadjan, R. Sayers, H. El Shinawi, C. Greaves, S. J. Skinner, Performance of $La_{2-x}Sr_xCo_{0.5}Ni_{0.5}O_{4\pm\delta}$ as an oxygen electrode for solid oxide reversible cells, Fuel Cells 11 (2011) 102–107, https://doi.org/10.1002/fuce.201000067.

[131] K. Yang, Y. Wang, L. Jiang, Y. Jin, Z. Yang, Cobalt-free perovskite $Ba_{0.95}La_{0.05}FeO_{3-\delta}$ as efficient and durable oxygen electrode for solid oxide electrolysis cells, Int. J. Hydrogen Energy 48 (2023) 27464–27472, https://doi.org/10.1016/j.ijhydene.2023.03.197.

[132] S. Kim, D.W. Joh, D.-Y. Lee, J. Lee, H.S. Kim, M.Z. Khan, J.E. Hong, S.-B. Lee, S. J. Park, R.-H. Song, M.T. Mehran, C.K. Rhee, T.-H. Lim, Microstructure tailoring of solid oxide electrolysis cell air electrode to boost performance and long-term durability, Chem. Eng. J. 410 (2021) 128318, https://doi.org/10.1016/j.cej.2020.128318.

[133] F. Chauveau, J. Mougin, J.M. Bassat, F. Mauvy, J.C. Grenier, A new anode material for solid oxide electrolyser: the neodymium nickelate $Nd_2NiO_{4+\delta}$, J. Power Sources 195 (2010) 744–749, https://doi.org/10.1016/j.jpowsour.2009.08.003.

[134] T. Chen, M. Liu, C. Yuan, Y. Zhou, X. Ye, Z. Zhan, C. Xia, S. Wang, High performance of intermediate temperature solid oxide electrolysis cells using $Nd_2NiO_{4+\delta}$ impregnated scandia stabilized zirconia oxygen electrode, J. Power Sources 276 (2015) 1–6, https://doi.org/10.1016/j.jpowsour.2014.11.042.

[135] A. Chroneos, D. Parfitt, J.A. Kilner, R.W. Grimes, Anisotropic oxygen diffusion in tetragonal $La_2NiO_{4+\delta}$: molecular dynamics calculations, J. Mater. Chem. 20 (2010) 266–270, https://doi.org/10.1039/B917118E.

[136] T. Ogier, J.M. Bassat, F. Mauvy, S. Fourcade, J.C. Grenier, K. Couturier, M. Petitjean, J. Mougin, Enhanced performances of structured oxygen electrodes for high temperature steam electrolysis, Fuel Cells 13 (2013) 536–541, https://doi.org/10.1002/fuce.201200201.

[137] D. Parfitt, A. Chroneos, A. Tarancón, J.A. Kilner, Oxygen ion diffusion in cation ordered/disordered $GdBaCo_2O_{5+\delta}$, J. Mater. Chem. 21 (2011) 2183–2186, https://doi.org/10.1039/C0JM02924F.

[138] Y.-H. Huang, R.I. Dass, Z.-L. Xing, J.B. Goodenough, Double perovskites as anode materials for solid-oxide fuel cells, Science 312 (2006) 254–257, https://doi.org/10.1126/science.1125877.

[139] E. Hernández, F. Baiutti, A. Morata, M. Torrell, A. Tarancón, Infiltrated mesoporous oxygen electrodes for high temperature co-electrolysis of H_2O and CO_2 in solid oxide electrolysis cells, J. Mater. Chem. A 6 (2018) 9699–9707, https://doi.org/10.1039/C8TA01045E.

[140] Y. Wang, T. Liu, S. Fang, F. Chen, Syngas production on a symmetrical solid oxide H_2O/CO_2 co-electrolysis cell with $Sr_2Fe_{1.5}Mo_{0.5}O_6$–$Sm_{0.2}Ce_{0.8}O_{1.9}$ electrodes, J. Power Sources 305 (2016) 240–248, https://doi.org/10.1016/j.jpowsour.2015.11.097.

[141] Y. Song, X. Zhang, Y. Zhou, Q. Jiang, F. Guan, H. Lv, G. Wang, X. Bao, Promoting oxygen evolution reaction by RuO_2 nanoparticles in solid oxide CO_2 electrolyzer, Energy Storage Mater. 13 (2018) 207–214, https://doi.org/10.1016/j.ensm.2018.01.013.

[142] Y. Zheng, J. Wang, B. Yu, W. Zhang, J. Chen, J. Qiao, J. Zhang, A review of high temperature co-electrolysis of H_2O and CO_2 to produce sustainable fuels using solid oxide electrolysis cells (SOECs): advanced materials and technology, Chem. Soc. Rev. 46 (2017) 1427–1463, https://doi.org/10.1039/C6CS00403B.

[143] P. Qiu, C. Li, B. Liu, D. Yan, J. Li, L. Jia, Materials of solid oxide electrolysis cells for H2O and CO2 electrolysis: a review, J. Adv. Ceram. 12 (2023) 1463–1510, https://doi.org/10.26599/JAC.2023.9220767.

[144] L. Yang, S. Wang, K. Blinn, M. Liu, Z. Liu, Z. Cheng, M. Liu, Enhanced Sulfur and coking tolerance of a mixed ion conductor for SOFCs: $BaZr_{0.1}Ce_{0.7}Y_{0.2-x}Yb_xO_{3-\delta}$, Science 326 (2009) 126–129, https://doi.org/10.1126/science.1174811.

[145] Y. Du, H. Ling, L. Zhao, H. Jiang, J. Kong, P. Liu, T. Zhou, The development of solid oxide electrolysis cells: critical materials, technologies and prospects, J. Power Sources 607 (2024) 234608, https://doi.org/10.1016/j.jpowsour.2024.234608.

[146] Y. Tian, N. Abhishek, C. Yang, R. Yang, S. Choi, B. Chi, J. Pu, Y. Ling, J.T.S. Irvine, G. Kim, Progress and potential for symmetrical solid oxide electrolysis cells, Matter 5 (2022) 482–514, https://doi.org/10.1016/j.matt.2021.11.013.

[147] Z. Gao, L.V. Mogni, E.C. Miller, J.G. Railsback, S.A. Barnett, A perspective on low-temperature solid oxide fuel cells, Energy Environ. Sci. 9 (2016) 1602–1644, https://doi.org/10.1039/C5EE03858H.

[148] G. Wang, D. Cui, Y. Yang, M. Cheng, Y. Ji, Ultra-high conductivity exploration and applications with multilayer heterostructure in solid oxide fuel cells: a review, J. Alloys Compd. 1004 (2024) 175742, https://doi.org/10.1016/j.jallcom.2024.175742.

[149] M. Wang, Y. Hua, Y. Gu, Y. Yin, L. Bi, High-entropy design in sintering aids for proton-conducting electrolytes of solid oxide fuel cells, Ceram. Int. 50 (2024) 4204–4212, https://doi.org/10.1016/j.ceramint.2023.11.104.

[150] G. Qin, J. Bao, J. Gao, F. Ruan, S. An, Z. Wang, L. Li, Enhanced grain boundary conductivity of Gd and Sc co-doping $BaZrO_3$ proton conductor for proton ceramic fuel cell, Chem. Eng. J. 466 (2023) 143114, https://doi.org/10.1016/j.cej.2023.143114.

[151] Z.U.D. Babar, M.B. Hanif, Y. Li, W.-T. Wang, H. Tian, C.-X. Li, Tailoring $BaCe_{0.7}Zr_{0.1}(Dy_{0.1}|Yb_{0.1})_{0.2}O_{3-}$ electrolyte through strategic Cu doping for low temperature proton conducting fuel cells: envisioned theoretically and experimentally, J. Energy Chem. 101 (2025) 692–701, https://doi.org/10.1016/j.jechem.2024.09.067.

[152] P. Chen, Z. Chen, D. Xue, D. Liu, W. Li, W. Chen, Q. Niu, M. Xu, W. Wang, C. Zhou, W. Zhou, Silicon-doped $BaZr_{0.1}Ce_{0.7}Y_{0.1}Yb_{0.1}O_{3-\delta}$ electrolyte with high mechanical strength and electrical performance for protonic ceramic cells, J. Mater. Sci. Technol. 243 (2026) 275–282, https://doi.org/10.1016/j.jmst.2025.03.104.

[153] M. Xiao, Z. Liu, H. Di, Y. Bai, G. Yang, D.A. Medvedev, Z. Luo, W. Wang, W. Zhou, R. Ran, Z. Shao, High-entropy materials for solid oxide cells: synthesis, applications, and prospects, J. Energy Chem. 104 (2025) 268–296, https://doi.org/10.1016/j.jechem.2024.12.009.

[154] T. He, P. Guan, L. Cong, Y. Ji, H. Sun, J. Wang, J. Liu, Assessment of performances of Ni–Cu–LSGM as anode materials for intermediate-temperature $LaGaO_3$-based solid oxide fuel cells, J. Alloys Compd. 393 (2005) 292–298, https://doi.org/10.1016/j.jallcom.2004.10.001.

[155] Y. Luo, X. Wu, Y. Shi, A.F. Ghoniem, N. Cai, Exergy analysis of an integrated solid oxide electrolysis cell-methanation reactor for renewable energy storage, Appl. Energy 215 (2018) 371–383, https://doi.org/10.1016/j.apenergy.2018.02.022.

[156] L. Lin, S. Chen, J. Quan, S. Liao, Y. Luo, C. Chen, C.-T. Au, Y. Shi, L. Jiang, Geometric synergy of steam/carbon dioxide Co-electrolysis and methanation in a tubular solid oxide electrolysis cell for direct power-to-methane, Energy Convers. Manag. 208 (2020) 112570, https://doi.org/10.1016/j.enconman.2020.112570.

[157] H. Zhang, Y. Zeng, R. Shi, D. Yan, L. Jia, W. Zhang, Evaluating the effect of flow configuration on the performance and thermal stress of solid oxide electrolysis cell, Mater. Sci. Eng. B 316 (2025) 118144, https://doi.org/10.1016/j.mseb.2025.118144.
[158] Y. Tu, J. Chai, S. Li, F. Han, Z. Zhang, W. Cai, The study of multiphysics field coupling and thermal stress in three types of Solid Oxide Electrolysis Cells (SOEC), Int. J. Electrochem. Sci. 19 (2024) 100789, https://doi.org/10.1016/j.ijoes.2024.100789.
[159] C. Liu, Z. Dang, G. Xi, Numerical study on thermal stress of solid oxide electrolyzer cell with various flow configurations, Appl. Energy 353 (2024) 122041, https://doi.org/10.1016/j.apenergy.2023.122041.
[160] Y. Kim, K. Lim, H. Salihi, S. Heo, H. Ju, The effects of stack configurations on the thermal management capabilities of solid oxide electrolysis cells, Energies 17 (2023) 125, https://doi.org/10.3390/en17010125.
[161] P. Pianko-Oprych, T. Zinko, Z. Jaworski, Simulation of thermal stresses for new designs of microtubular solid oxide fuel cell stack, Int. J. Hydrogen Energy 40 (2015) 14584–14595, https://doi.org/10.1016/j.ijhydene.2015.05.164.
[162] M. Dillig, J. Leimert, J. Karl, Planar high temperature heat pipes for SOFC/SOEC stack applications, Fuel Cells 14 (2014) 479–488, https://doi.org/10.1002/fuce.201300224.
[163] Y. Sun, X. Hu, J. Gao, Y. Han, A. Sun, N. Zheng, W. Shuai, G. Xiao, M. Guo, M. Ni, H. Xu, Solid oxide electrolysis cell under real fluctuating power supply with a focus on thermal stress analysis, Energy 261 (2022) 125096, https://doi.org/10.1016/j.energy.2022.125096.
[164] C. Cheng, W. Ding, J. Shen, P. Liao, C. Yu, B. Miao, Y. Zhou, H. Li, H. Zhang, Z. Zhong, Multi-physics coupling simulation of H_2O–CO_2 Co-Electrolysis using flat tubular solid oxide electrolysis cells, Processes 13 (2025) 3192, https://doi.org/10.3390/pr13103192.
[165] Z. Zhang, C. Guan, L. Xie, J.-Q. Wang, Design and analysis of a novel opposite trapezoidal flow channel for solid oxide electrolysis cell stack, Energies 16 (2022) 159, https://doi.org/10.3390/en16010159.
[166] Y. Tu, Z. Zhang, H. Lin, W. Cai, Numerical investigation of a novel design for an elliptical channel solid oxide electrolysis cell for CO_2/H_2O Co-Electrolysis, Fuel 385 (2025) 134095, https://doi.org/10.1016/j.fuel.2024.134095.
[167] W. Kong, Z. Han, S. Lu, X. Gao, X. Wang, A novel interconnector design of SOFC, Int. J. Hydrogen Energy 45 (2020) 20329–20338, https://doi.org/10.1016/j.ijhydene.2019.10.252.
[168] M. Guo, Q. He, C. Cheng, D. Zhao, M. Ni, New interconnector designs for electrical performance enhancement of solid oxide fuel cells: a 3D modelling study, J. Power Sources 533 (2022) 231373, https://doi.org/10.1016/j.jpowsour.2022.231373.
[169] L. Wang, M. Rao, S. Diethelm, T.-E. Lin, H. Zhang, A. Hagen, F. Maréchal, J. Van Herle, Power-to-methane via co-electrolysis of H_2O and CO_2: the effects of pressurized operation and internal methanation, Appl. Energy 250 (2019) 1432–1445, https://doi.org/10.1016/j.apenergy.2019.05.098.
[170] L. Wang, M. Pérez-Fortes, H. Madi, S. Diethelm, J.V. Herle, F. Maréchal, Optimal design of solid-oxide electrolyzer based power-to-methane systems: a comprehensive comparison between steam electrolysis and co-electrolysis, Appl. Energy 211 (2018) 1060–1079, https://doi.org/10.1016/j.apenergy.2017.11.050.
[171] C. Wang, Z. Li, S. Zhao, L. Xia, M. Zhu, M. Han, M. Ni, Modelling of an integrated protonic ceramic electrolyzer cell (PCEC) for methanol synthesis, J. Power Sources 559 (2023) 232667, https://doi.org/10.1016/j.jpowsour.2023.232667.
[172] F. Lonis, V. Tola, G. Cau, Assessment of integrated energy systems for the production and use of renewable methanol by water electrolysis and CO_2 hydrogenation, Fuel 285 (2021) 119160, https://doi.org/10.1016/j.fuel.2020.119160.
[173] M. Qi, J. Lee, S. Hong, J. Kim, Y. Liu, J. Park, I. Moon, Flexible and efficient renewable-power-to-methane concept enabled by liquid CO_2 energy storage: optimization with power allocation and storage sizing, Energy 256 (2022) 124583, https://doi.org/10.1016/j.energy.2022.124583.
[174] N. Zheng, J. Zhu, H. Zhu, J. Xuan, H. Xu, M. Ni, Toward a circular carbon economy: production of green C1 compounds through high-temperature CO_2 electrolysis, Nexus 2 (2025) 100053, https://doi.org/10.1016/j.ynexs.2024.100053.
[175] J. Haider, B. Lee, C. Choe, M. Abdul Qyyum, S. Shiung Lam, H. Lim, SNG production with net zero outflow of CO_2 in an integrated energy system: an energy and economic aspects, Energy Convers. Manag. 270 (2022) 116167, https://doi.org/10.1016/j.enconman.2022.116167.
[176] P. Guo, N.B. Kumar, Y. Elmasry, A. Alanazi, T.I. Alanazi, A. Armghan, A. M. Algelany, M. Wae-hayee, CO_2 hydrogenation for geothermal energy storage through synthetic natural gas production and byproduct of refrigeration and freshwater using solid oxide electrolyzer cell (SOEC) and methanation reactor; techno-economic evaluation and multi-objective optimization, J. CO2 Util. 69 (2023) 102395, https://doi.org/10.1016/j.jcou.2023.102395.
[177] R. Jacobs, J. Liu, H. Abernathy, D. Morgan, Machine learning design of perovskite catalytic properties, Adv. Energy Mater. 14 (2024) 2303684, https://doi.org/10.1002/aenm.202303684.